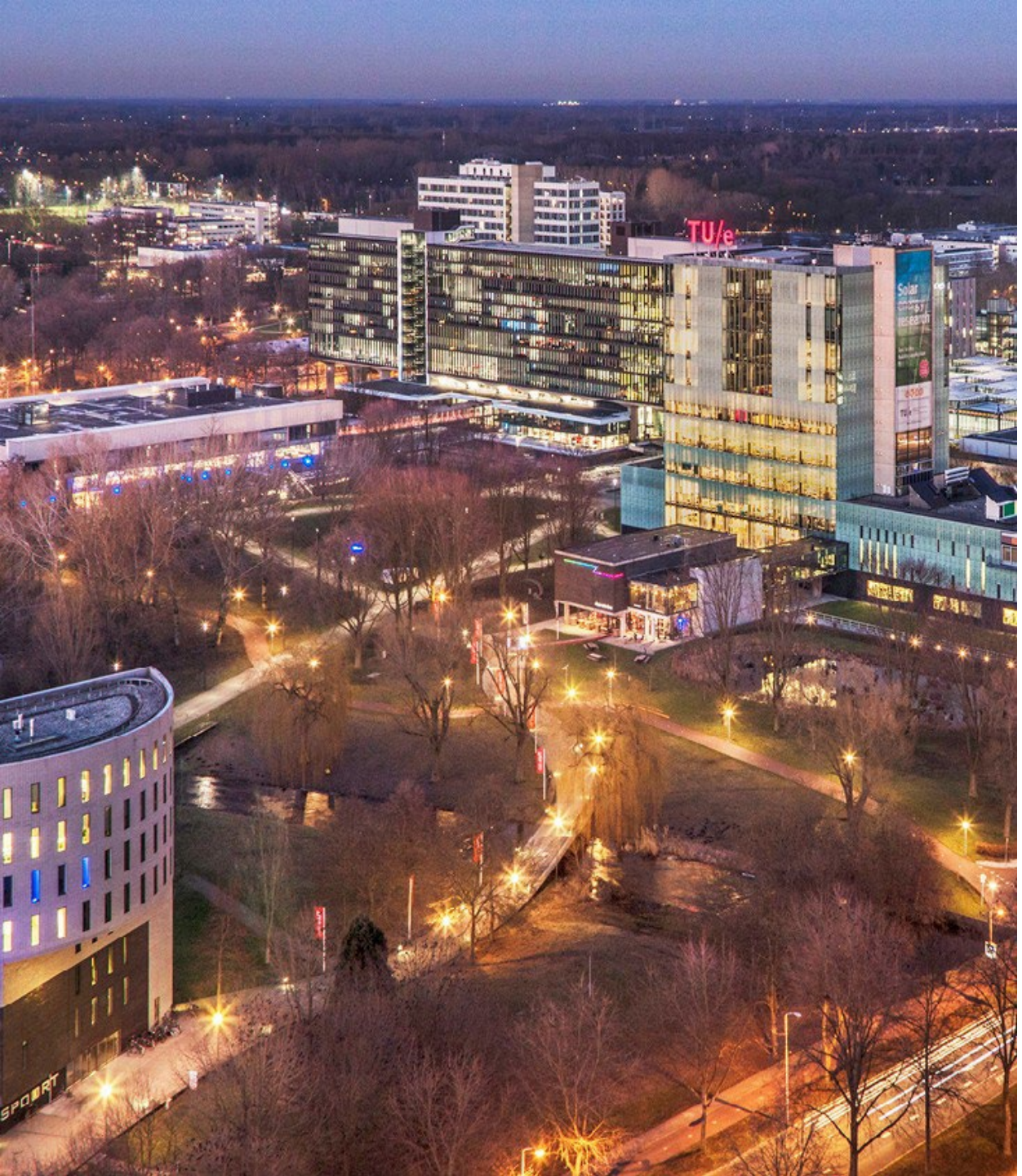
From omics data to networks to mechanistic models

EMBL-EBI Course

Systems biology: from large datasets to biological insight

15-05-2026

Bi-rong Wang (PhD Candidate at Biomedical Engineering, TU/e)



TU/e BmE

Eindhoven, University of Technology

Biomedical Engineering

SysBio Oncology

Systems Biology for Oncology

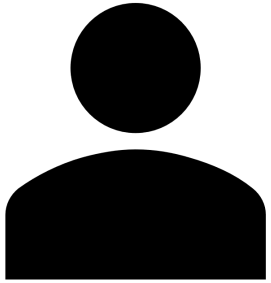
UK HD

Internship at Saezlab, Heidelberg

Common research topic

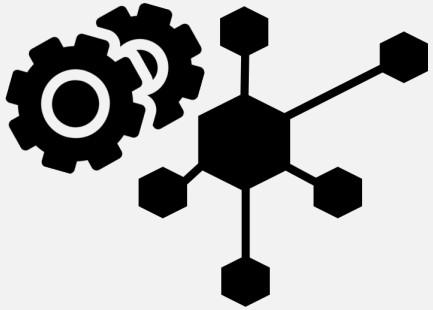
Unraveling signaling dynamics through computational models

Anti-cancer treatment



Patient

Therapy response



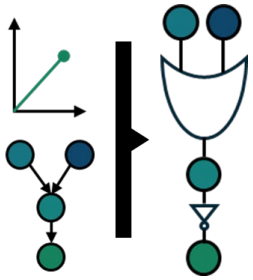
Mechanistic hypotheses

Regulation of proteins within signaling pathways

What type of computational models?

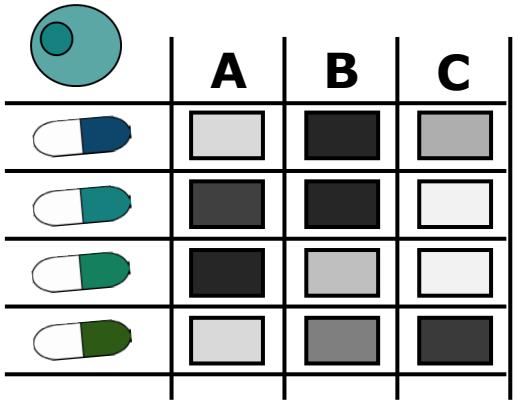
What type of data?

What type of computational models?



CellNOpt
(Terfve et al., 2012)

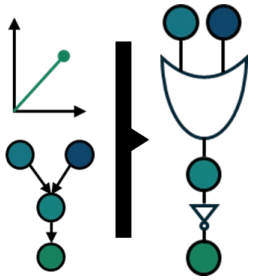
What type of data?



Perturbation data

Limitations with patient data

What type of computational models?



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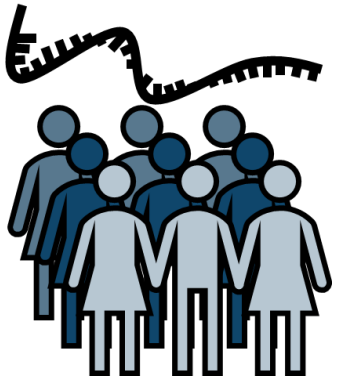
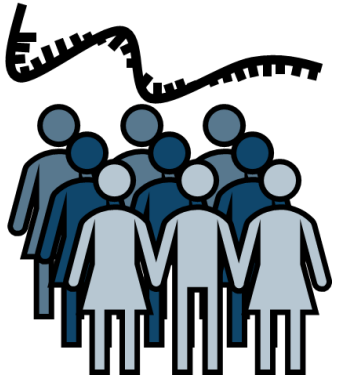
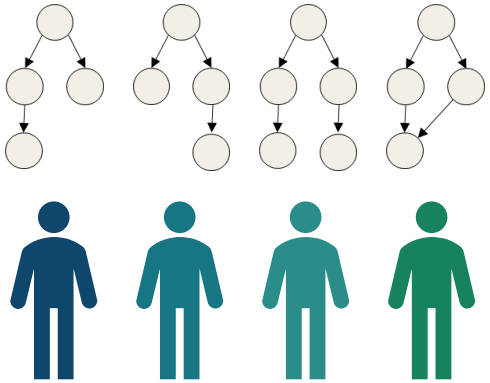


Omics data

Omics dataset

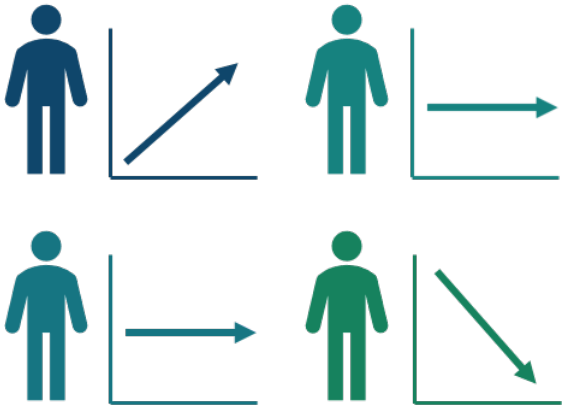


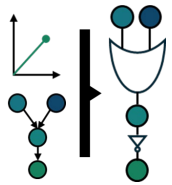
Static networks



COSMOS (Dugourd et al., 2024)

Dynamic models





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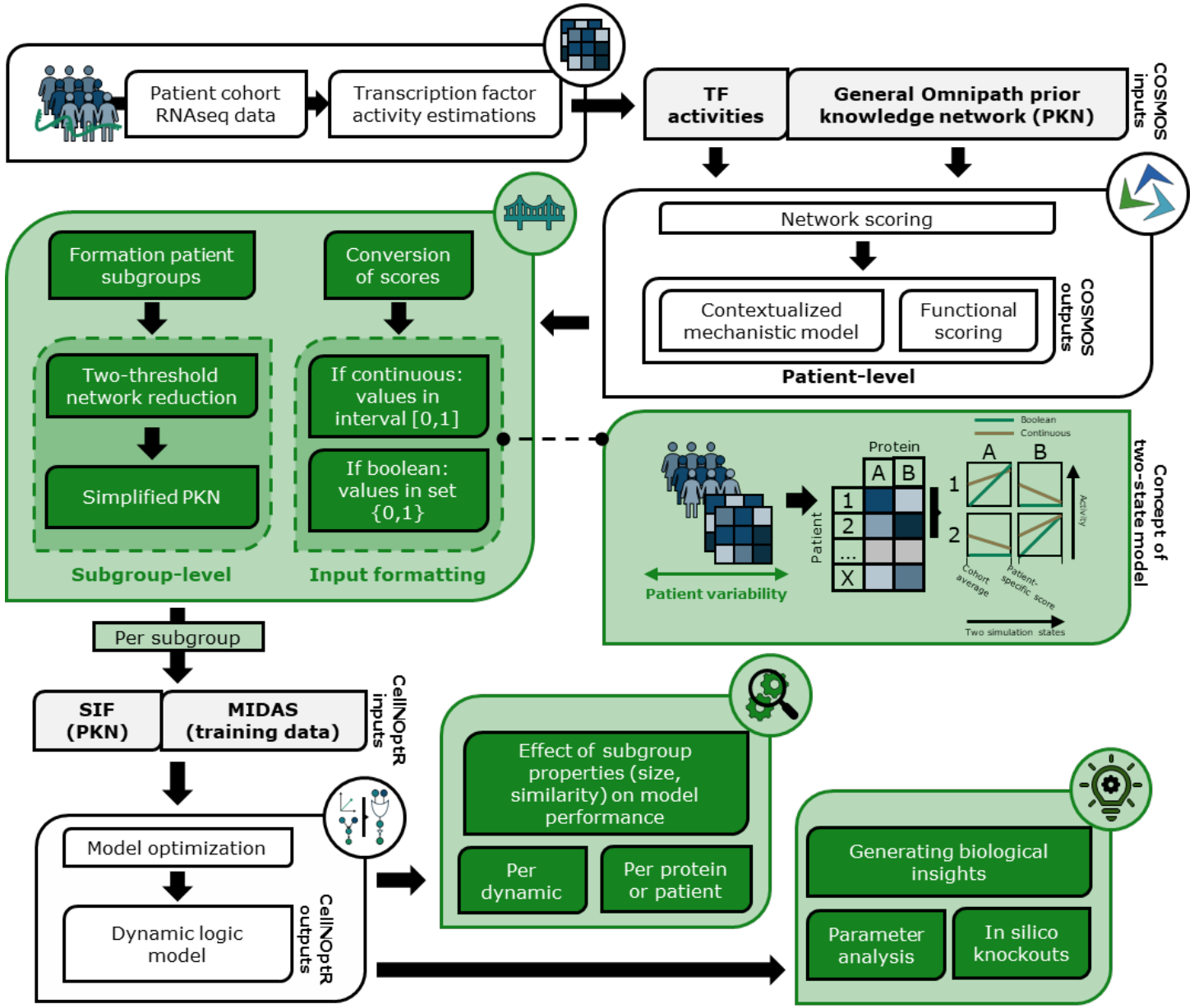
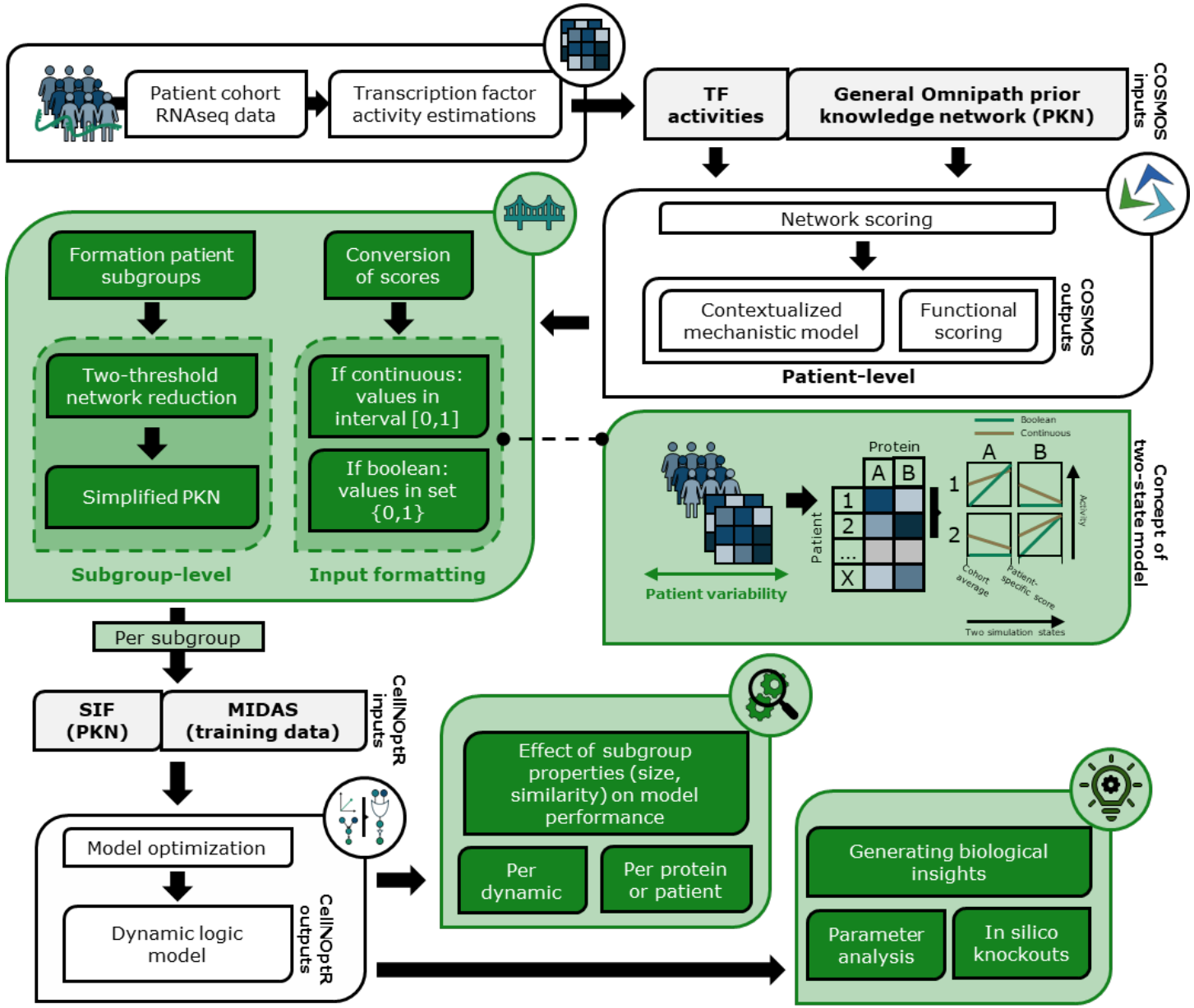


Functional **I**ntegration of **C**ontextualized omics to **U**nravel signaling dynamic**S**


Patient cohort RNAseq data
Transcription factor activity estimations
TF activities
General Omnipath prior knowledge network (PKN)
COSMOS inputs
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1
2
...
X
Activity
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Two simulation states
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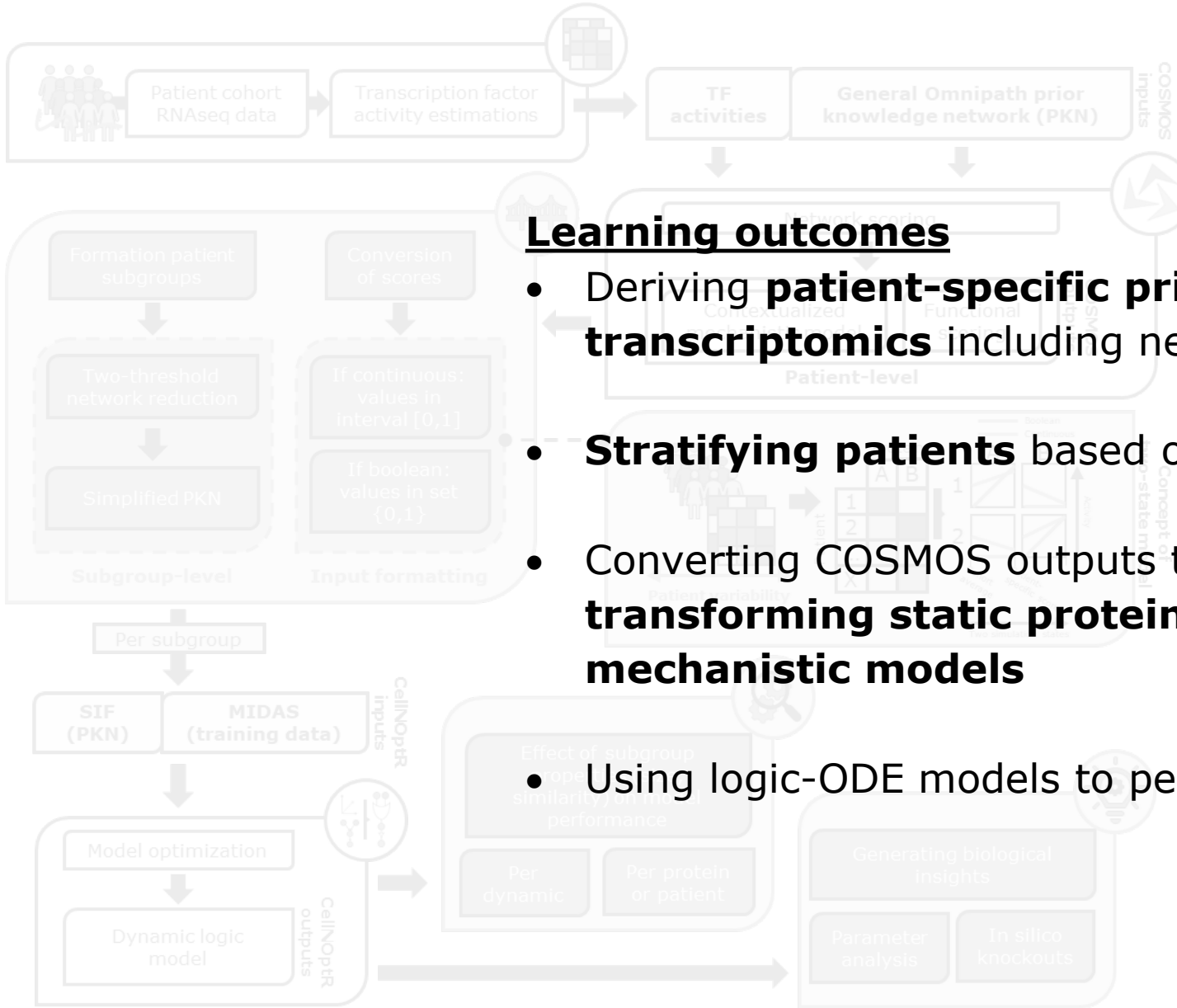
Effect of subgroup properties (size, similarity) on model performance
Per dynamic
Per protein or patient
Generating biological insights
Parameter analysis
In silico knockouts

FICUS



Learning outcomes

- Deriving **patient-specific prior knowledge networks** from **bulk transcriptomics** including network simplification
- **Stratifying patients** based on protein networks
- Converting COSMOS outputs to CellNOpt compatible inputs, **transforming static protein representations into dynamic mechanistic models**
- Using logic-ODE models to perform **in silico knockout screenings**

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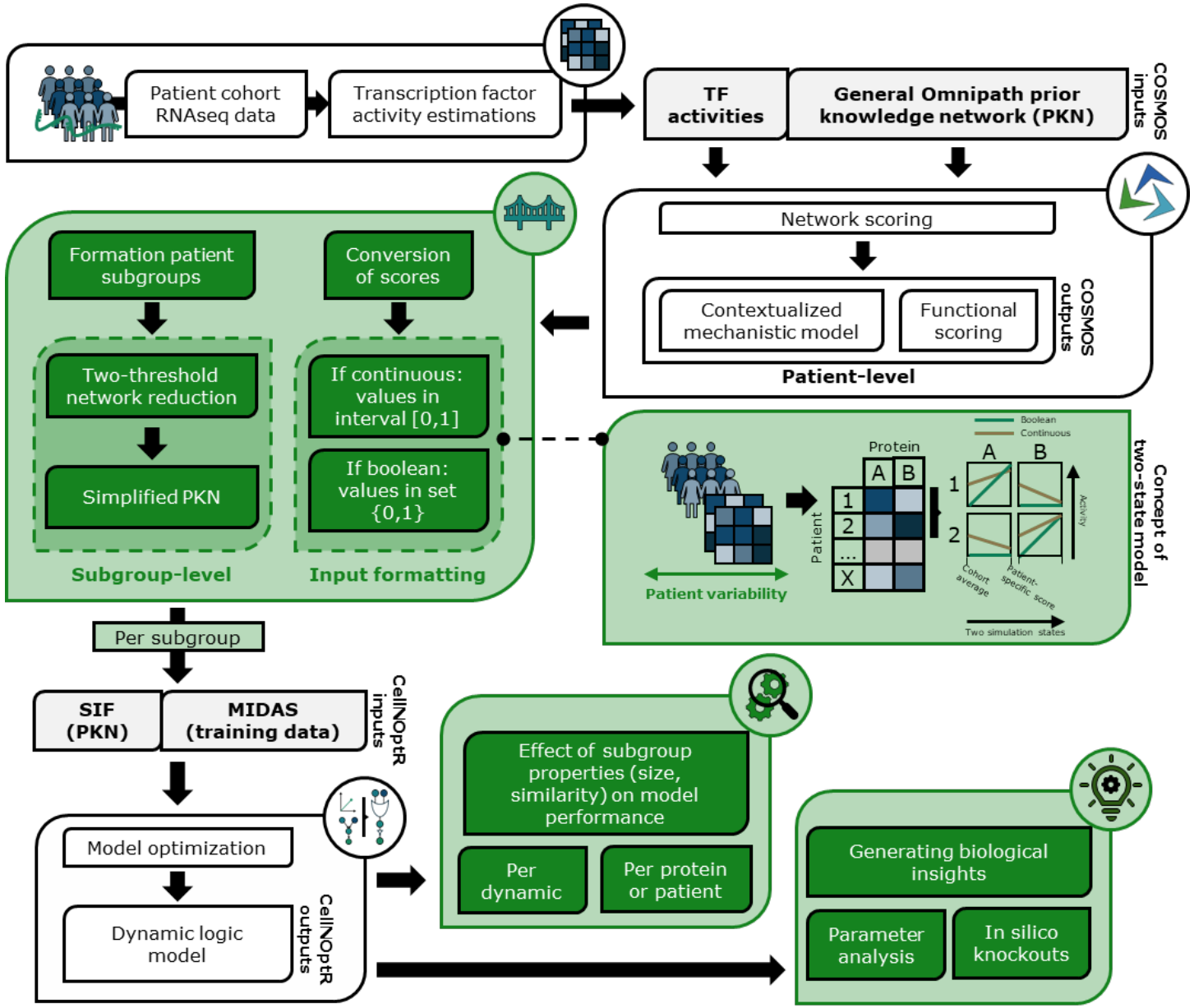
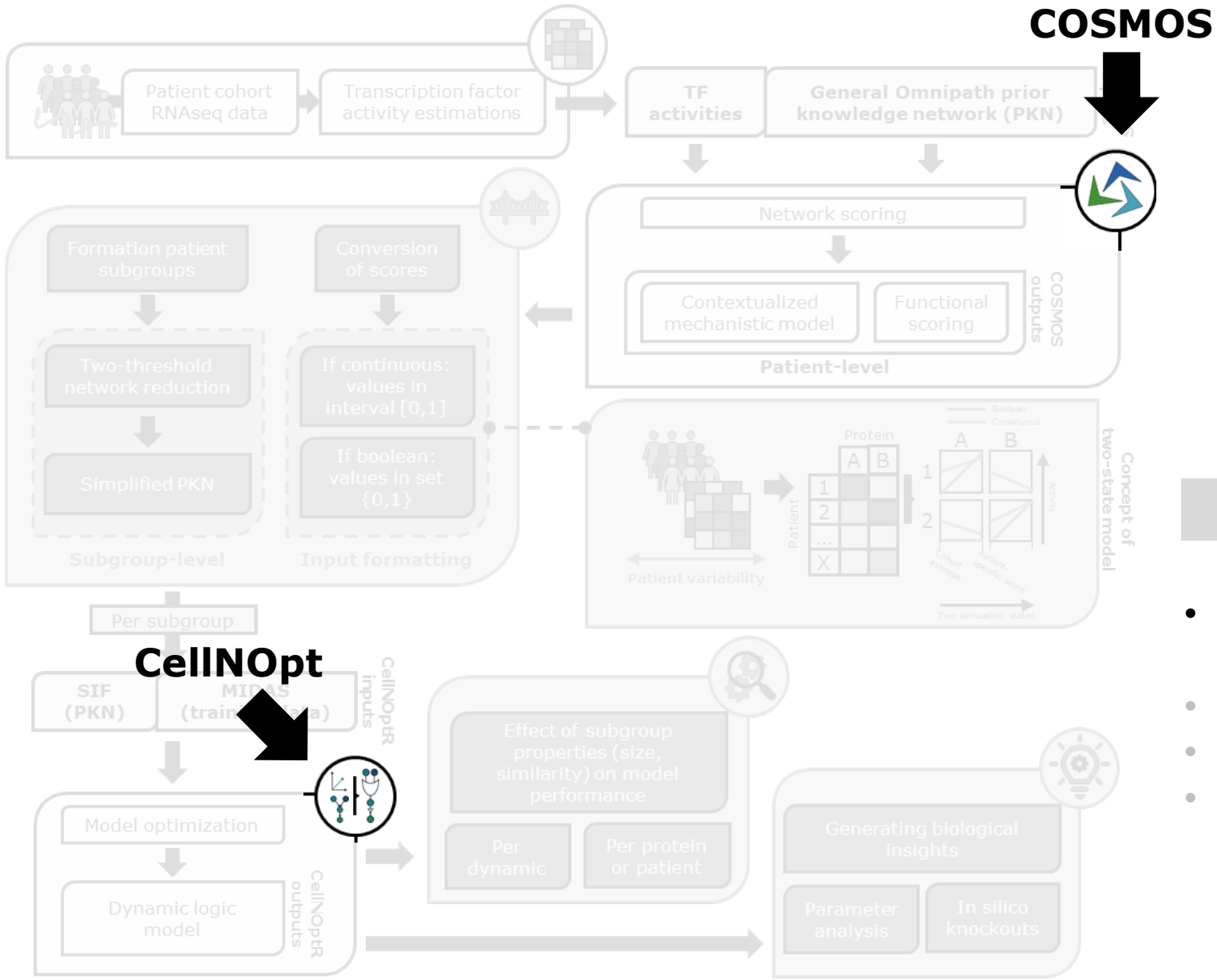


FICUS

- Overview of the main tools (COSMOS, CellNOpt)
- Preparation of PKN
- Preparation of training data
- Post-hoc analyses



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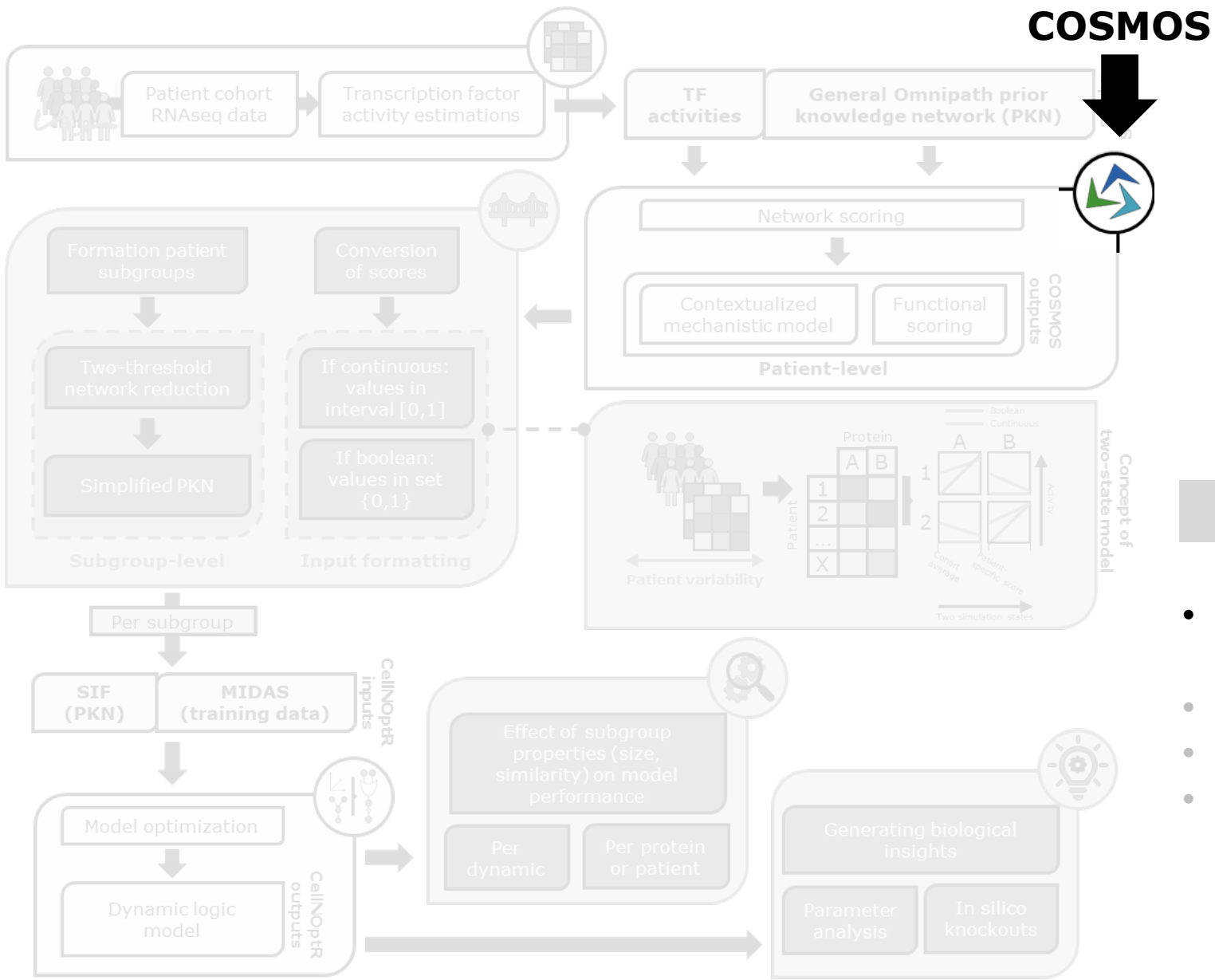

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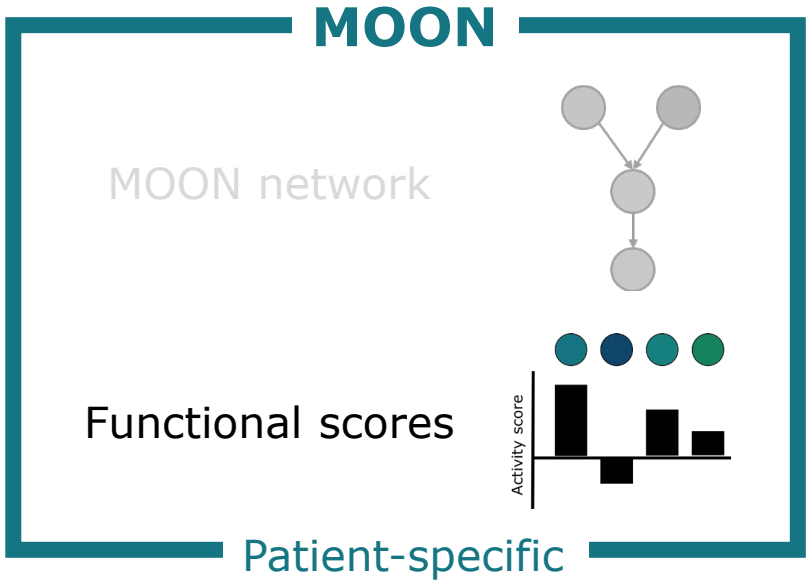
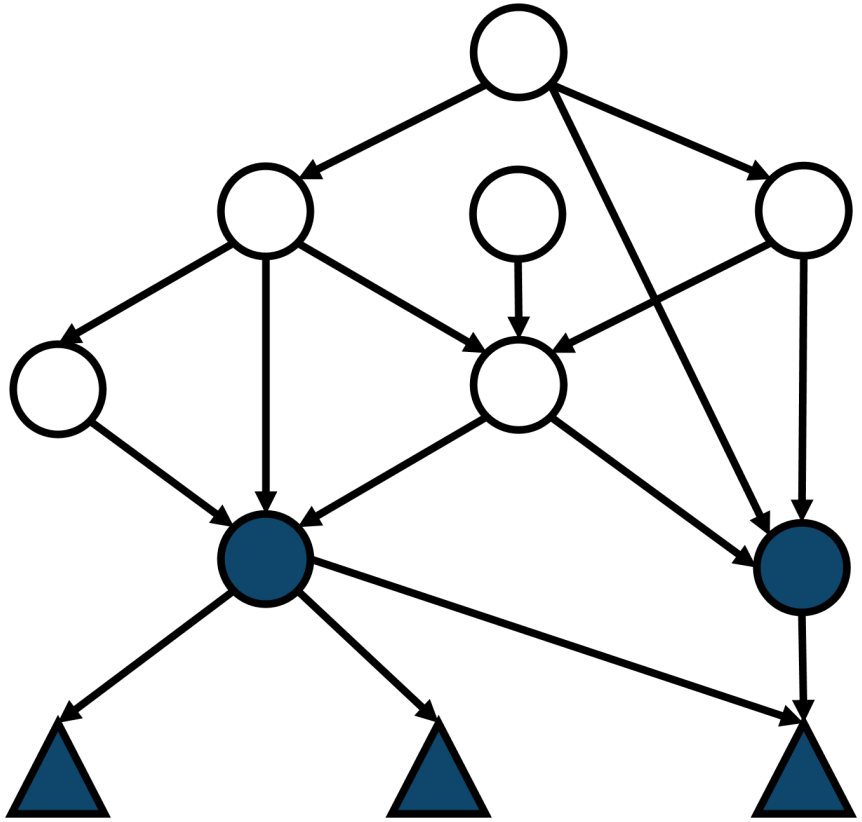
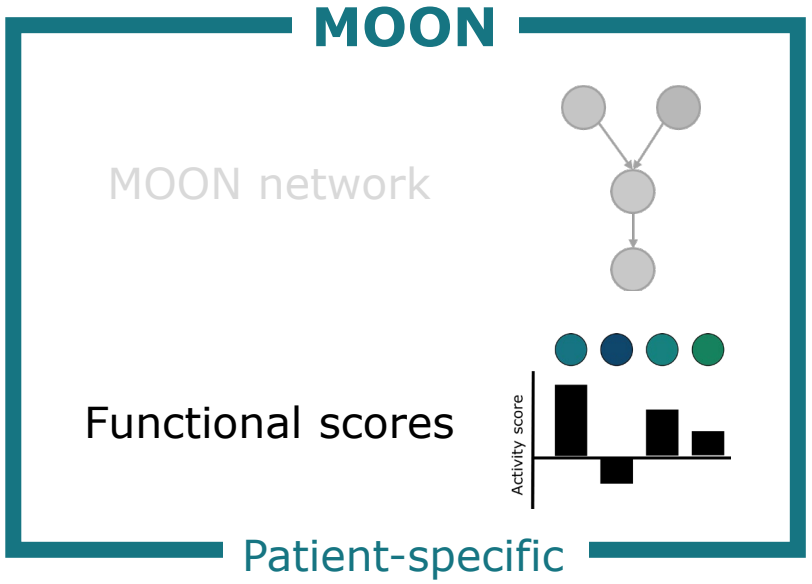
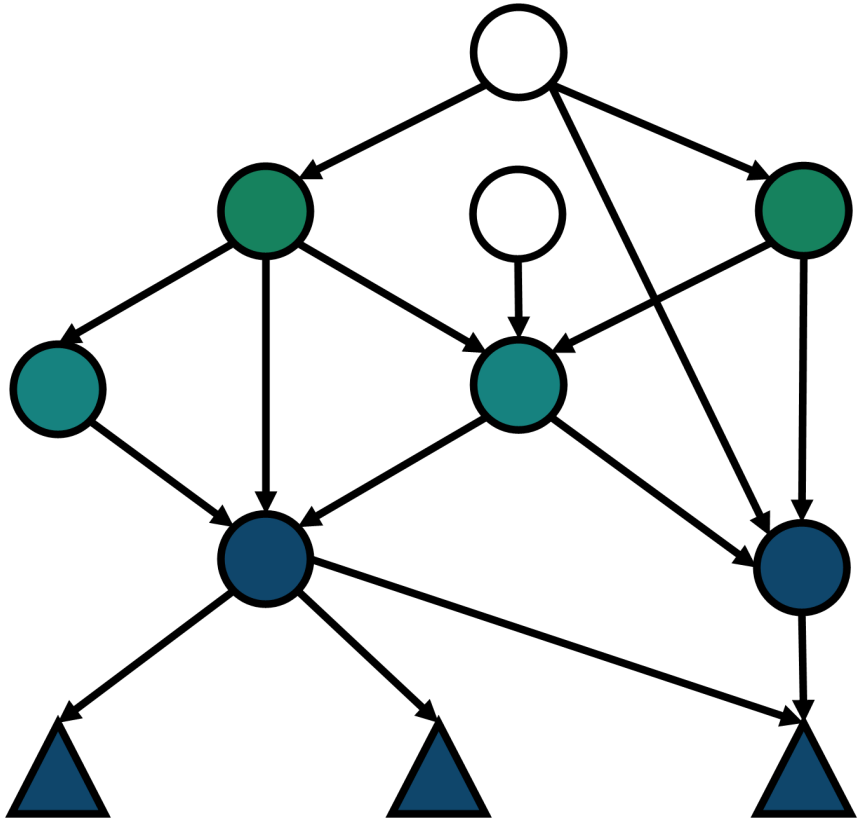




1) General network

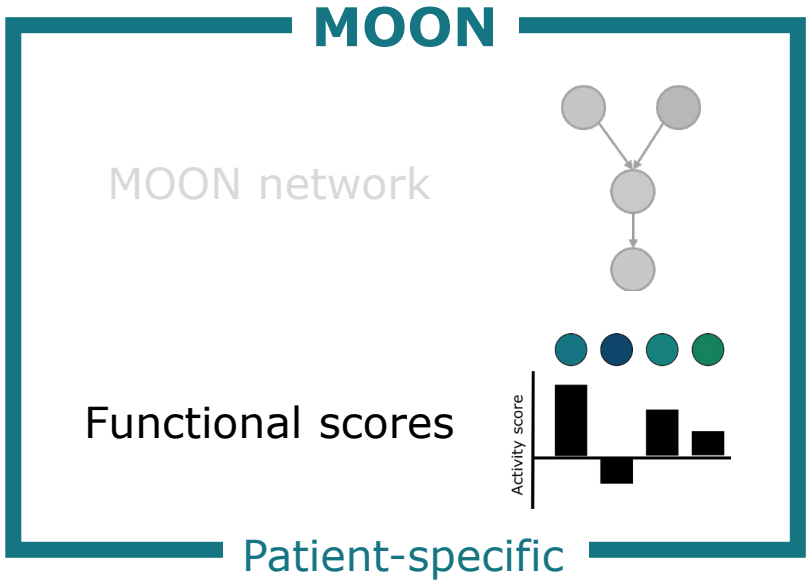
COSMOS (Dugourd et al., 2024)

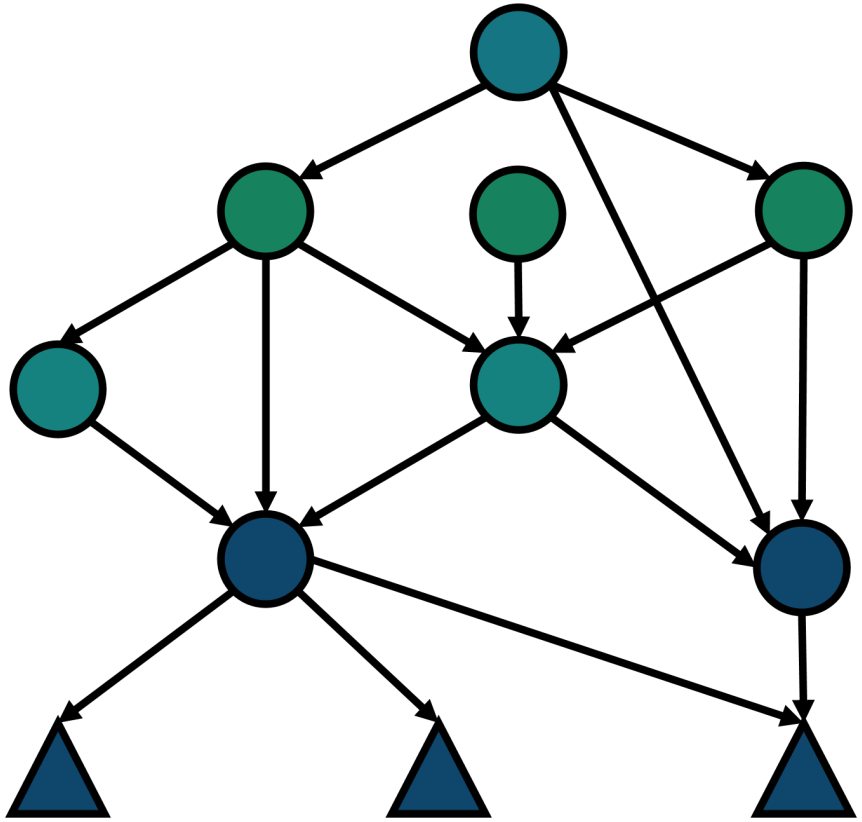




2) Estimate upstream protein activities

COSMOS (Dugourd et al., 2024)





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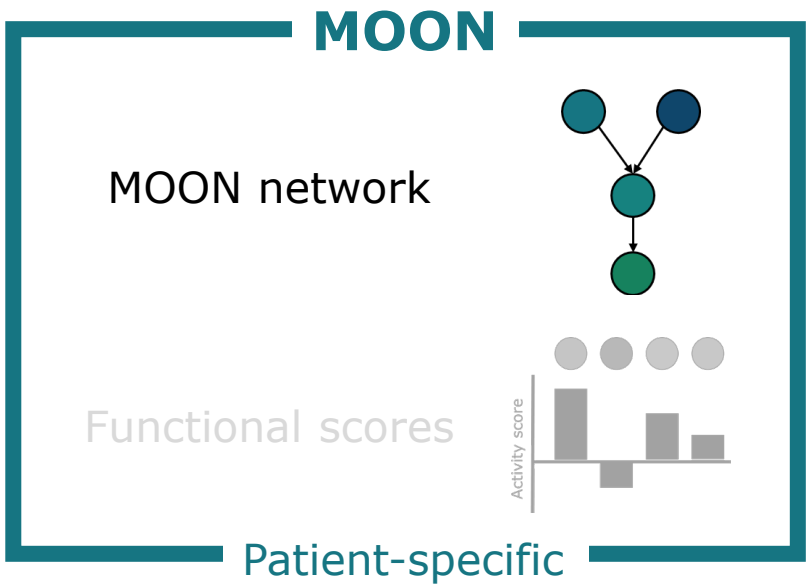
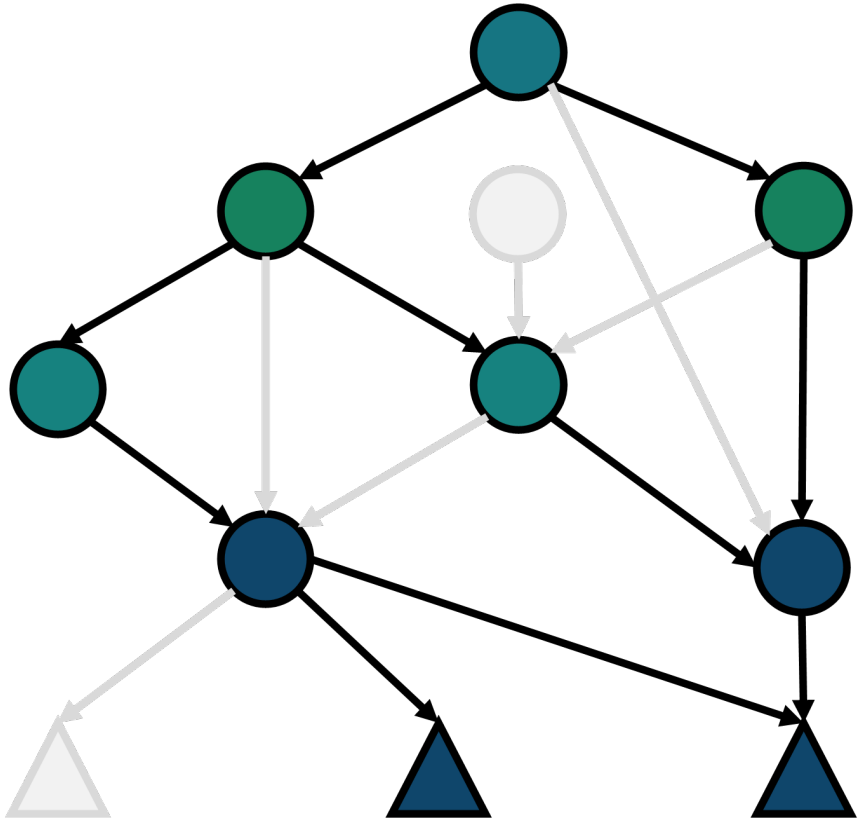
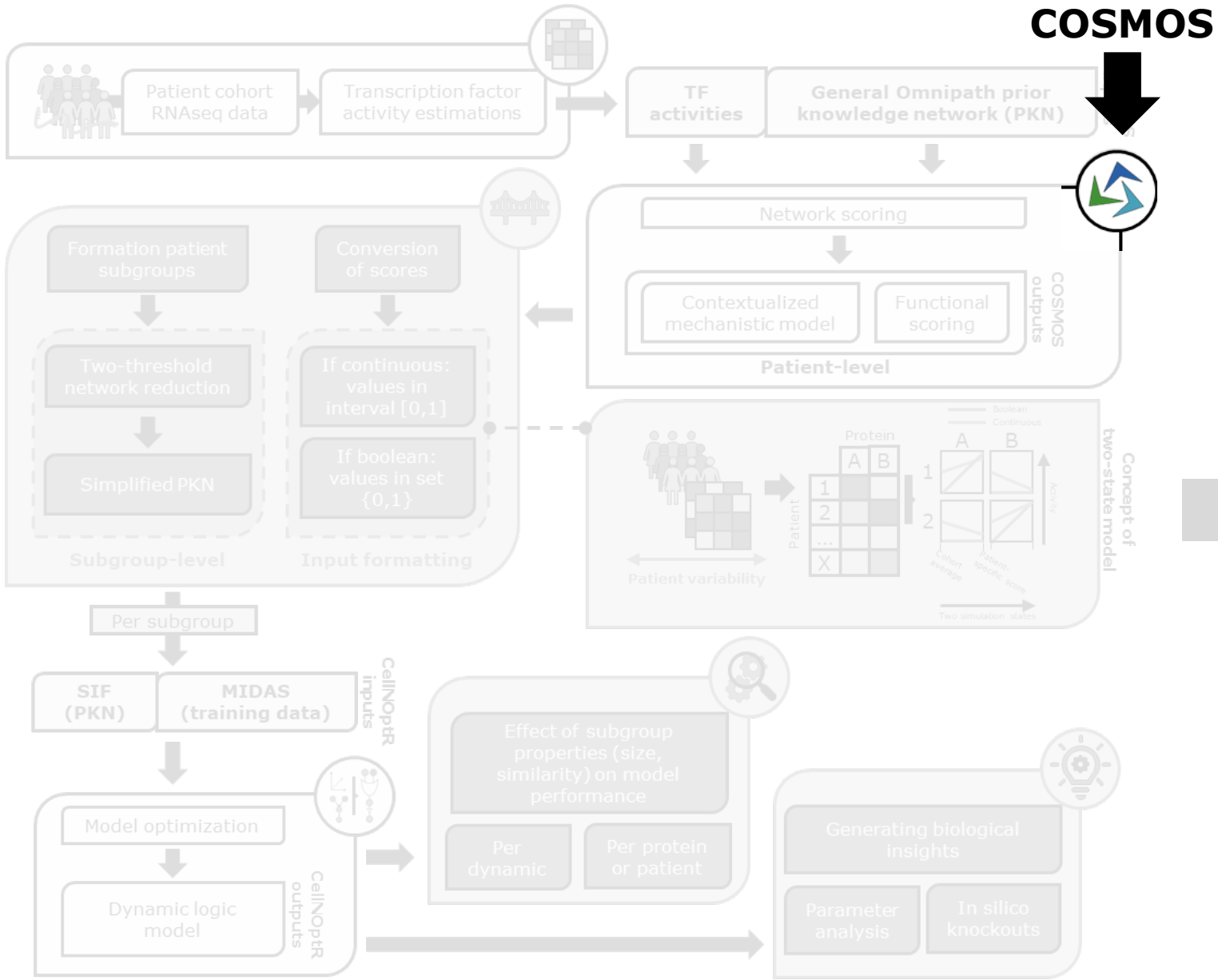




3) Adjust network to get sample-specific MOON network

COSMOS (Dugourd et al., 2024)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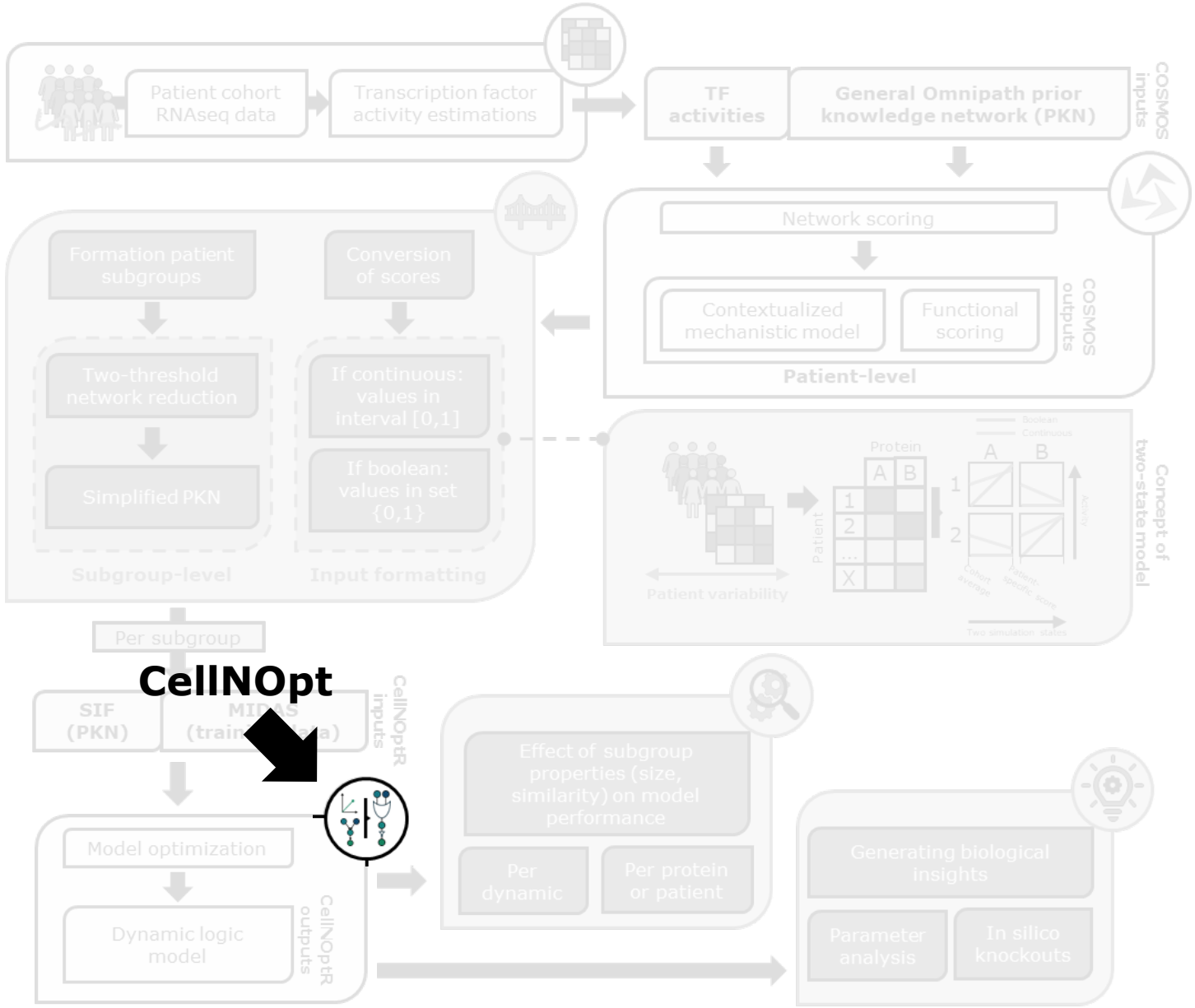
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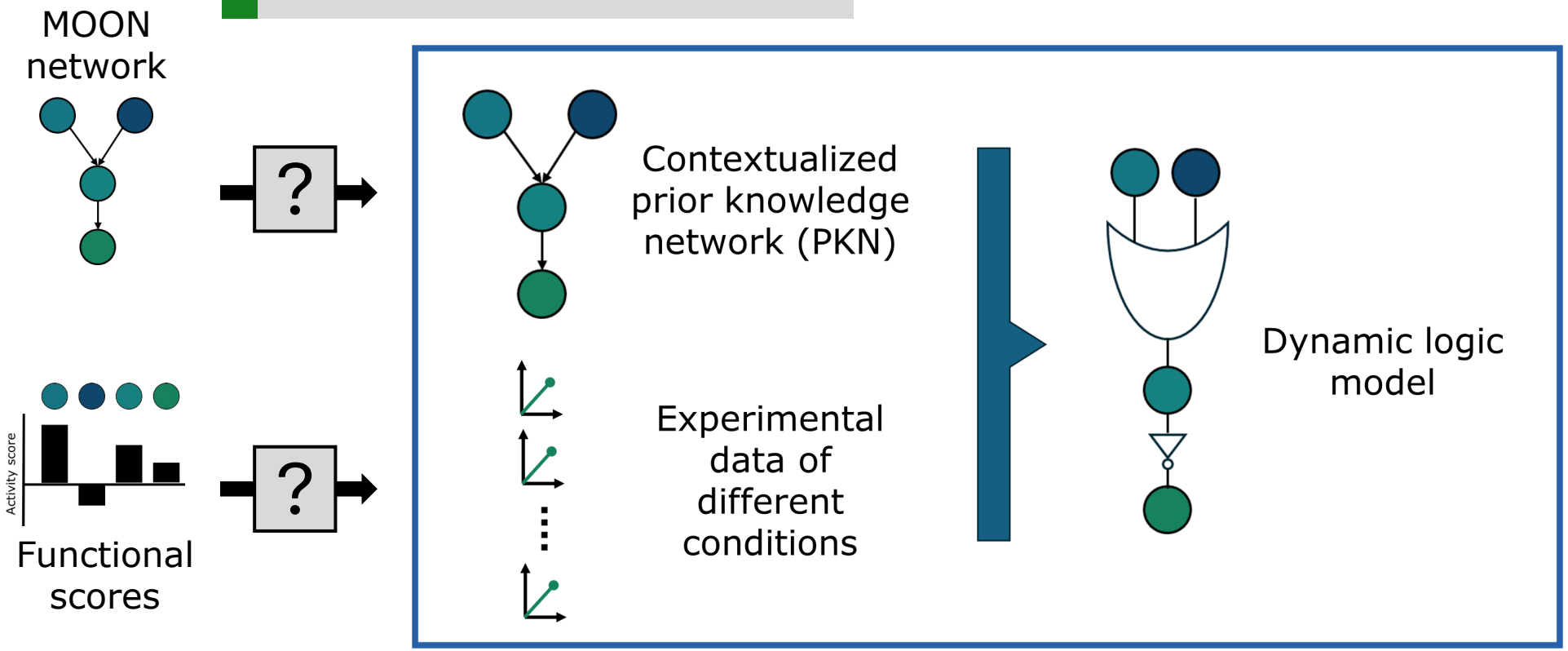
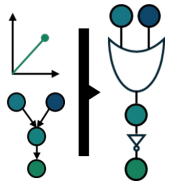
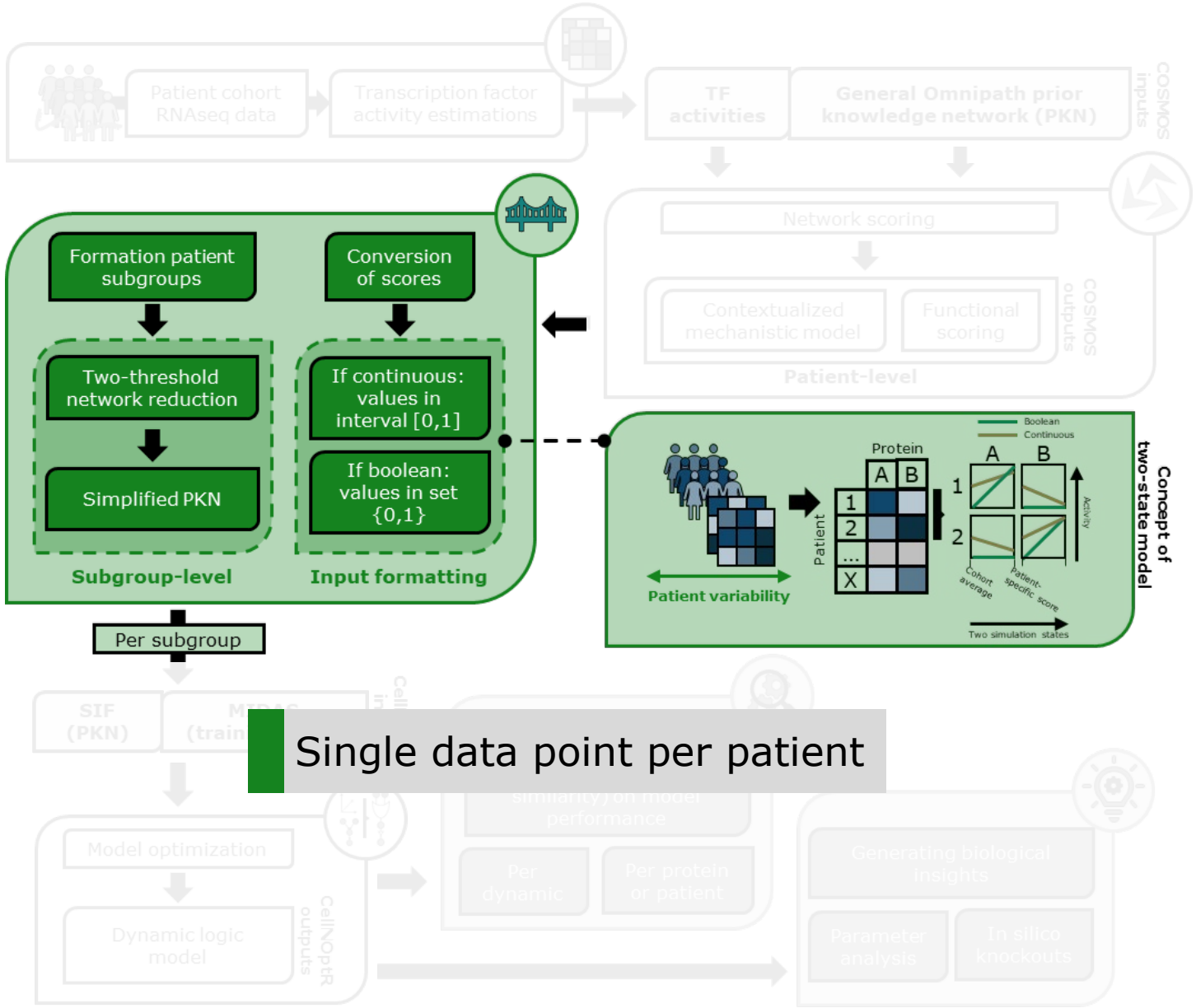
FICUS

Single data point per patient





CellNOpt (Terfve et al., 2012)



Single data point per patient

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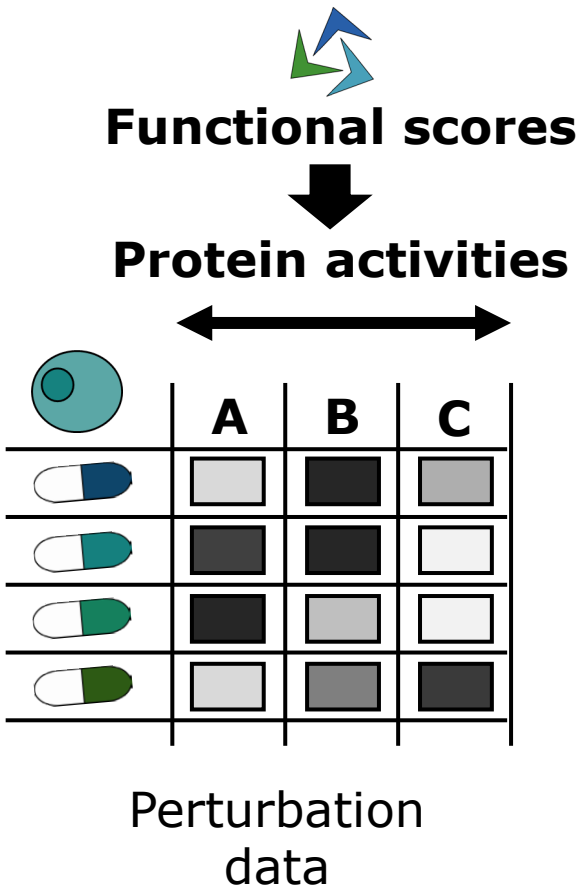
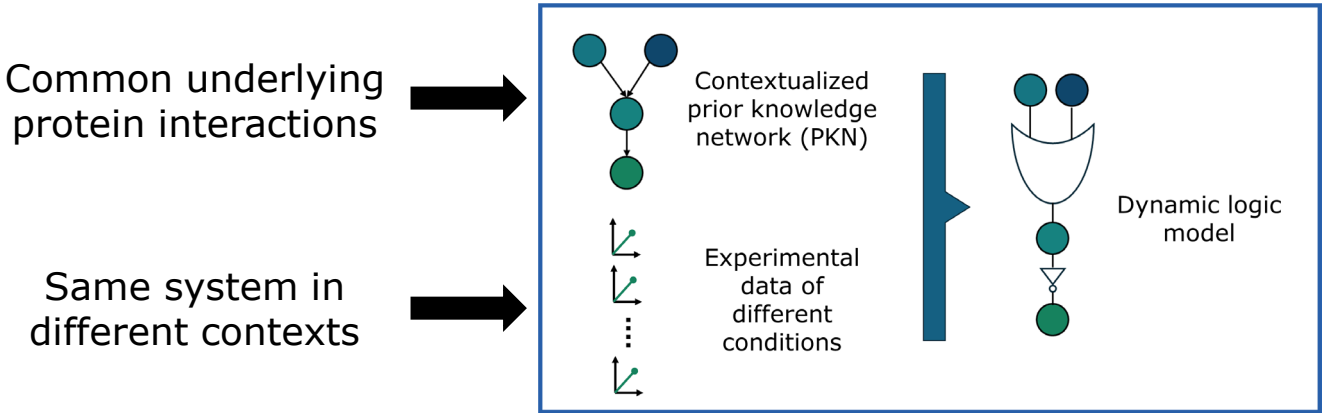
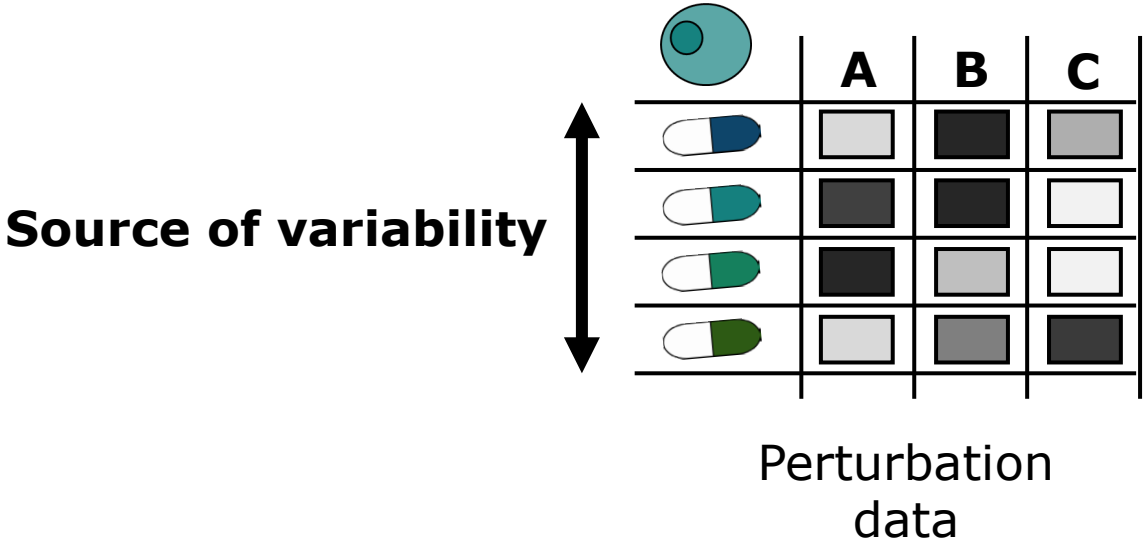

Functional scores
Protein activities
A
B
C
Perturbation data


Common underlying protein interactions
Same system in different contexts
Contextualized prior knowledge network (PKN)
Experimental data of different conditions
Dynamic logic model


A
B
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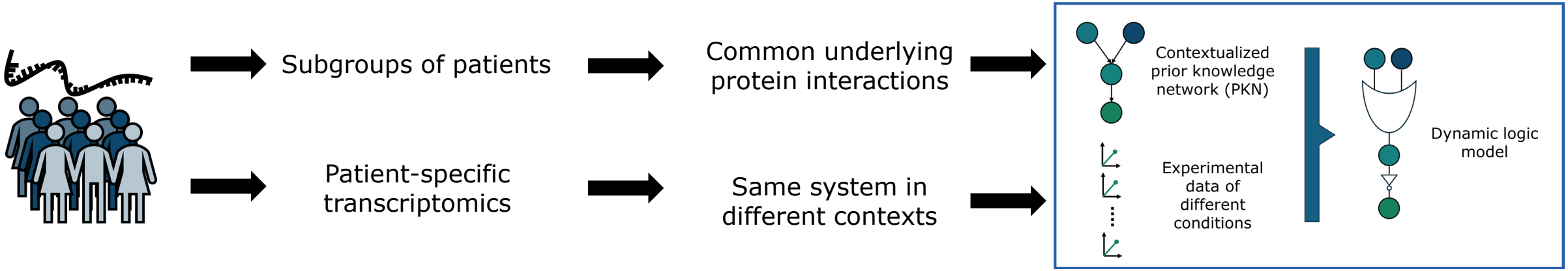
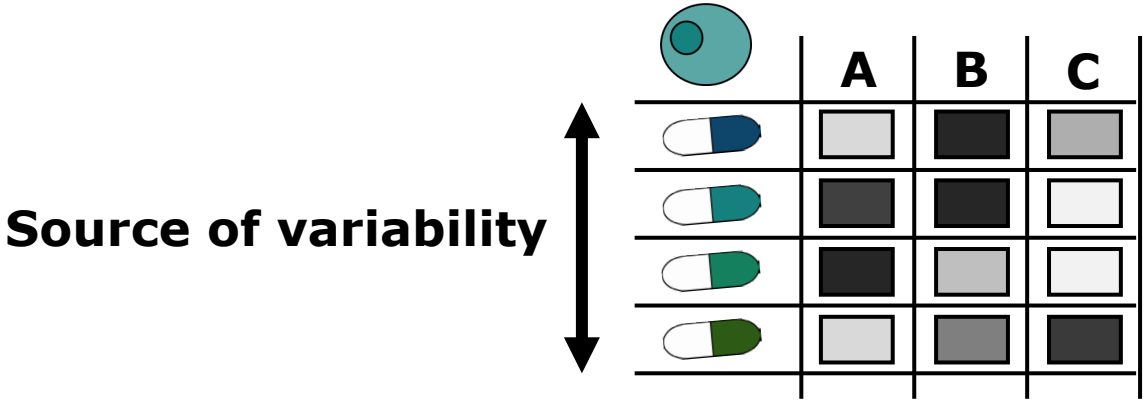
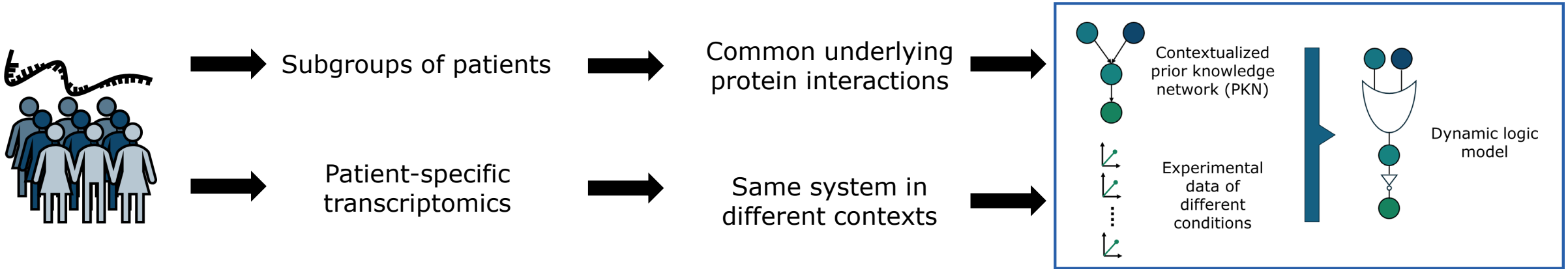
Source of variability
Perturbation data





Perturbation data


Subgroups of patients
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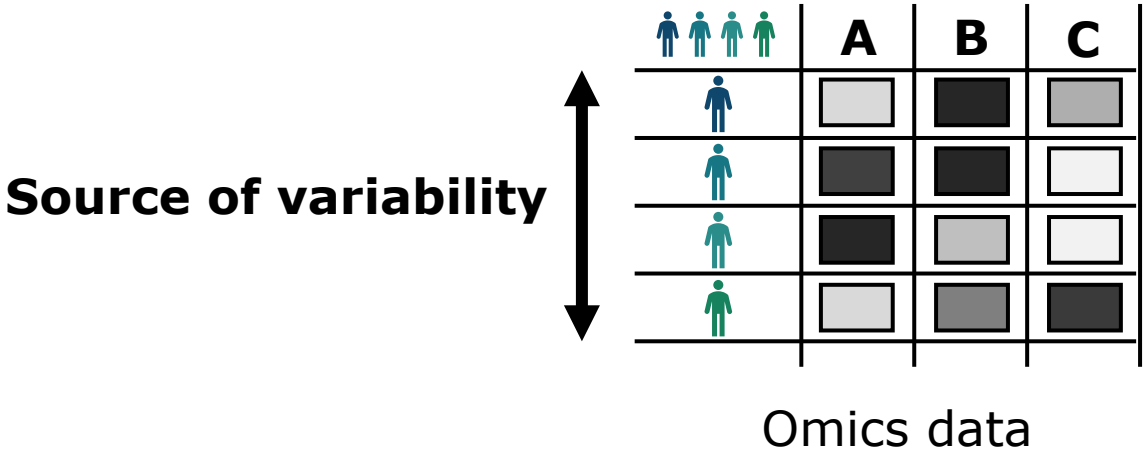
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A
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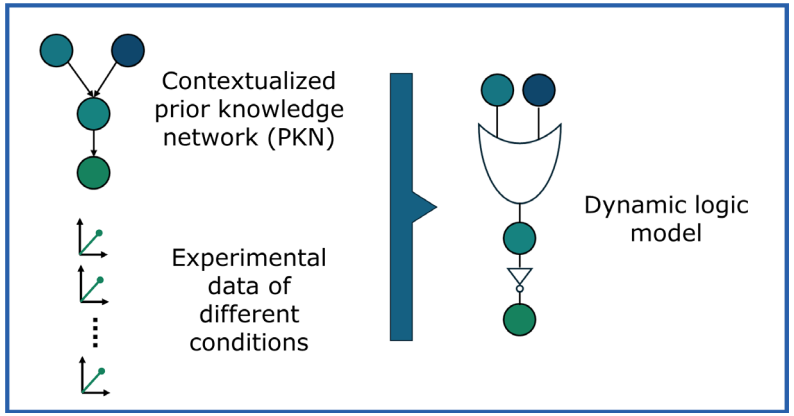
Omics data

Subgroups of patients → Common underlying protein interactions →

Patient-specific transcriptomics → Same system in different contexts →



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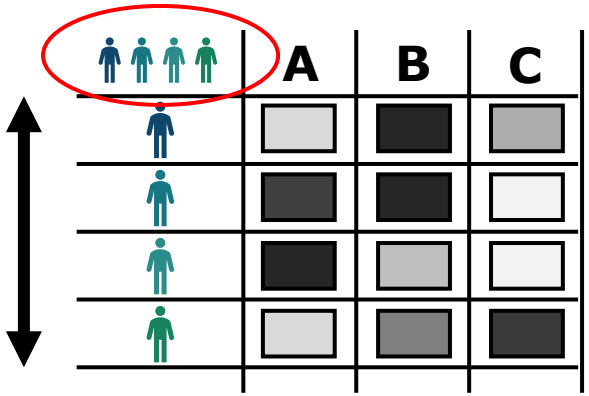

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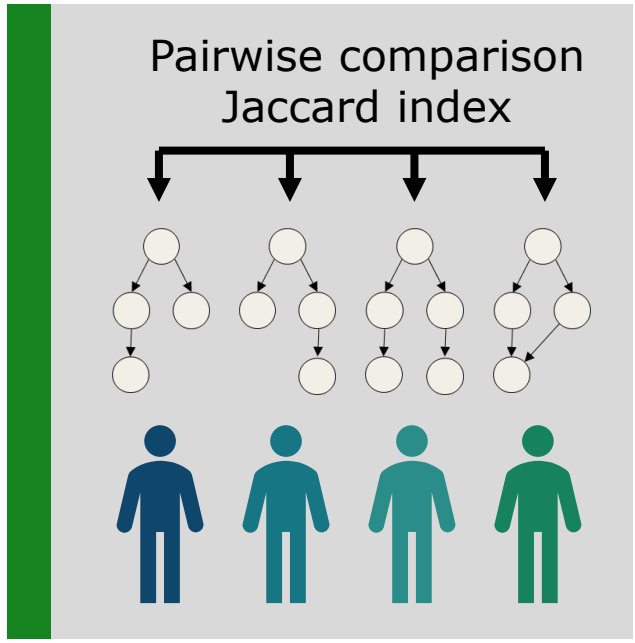
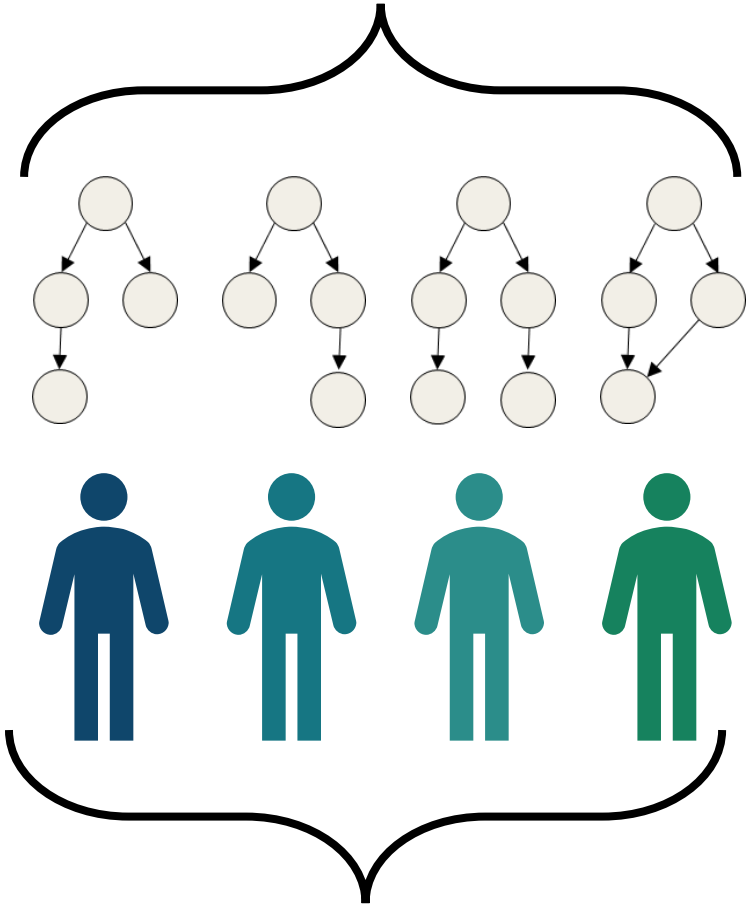


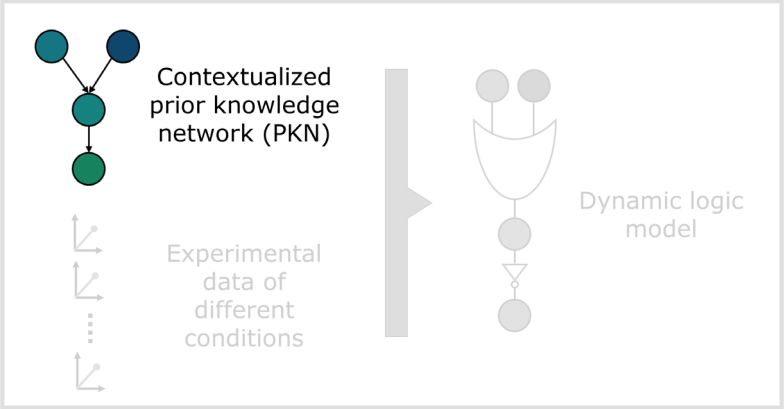
Subgroup formation

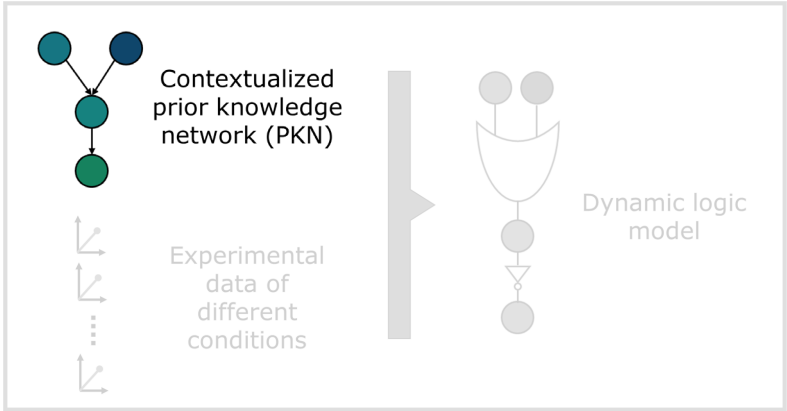
Hierarchical clustering

PKN : union of patient-specific MOON networks



Subgroup



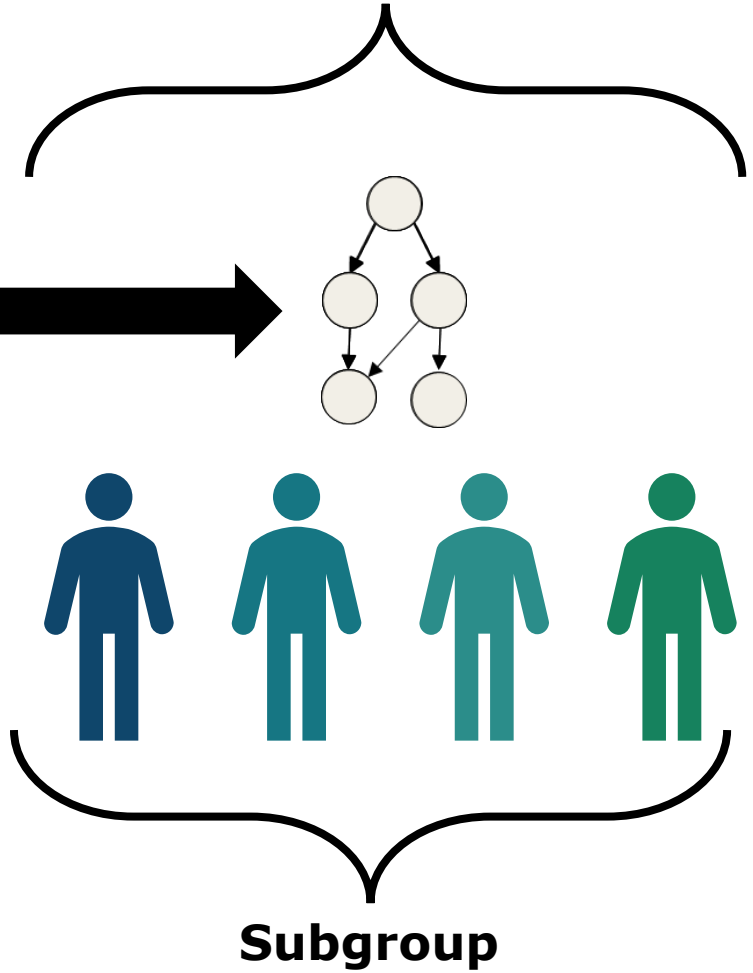


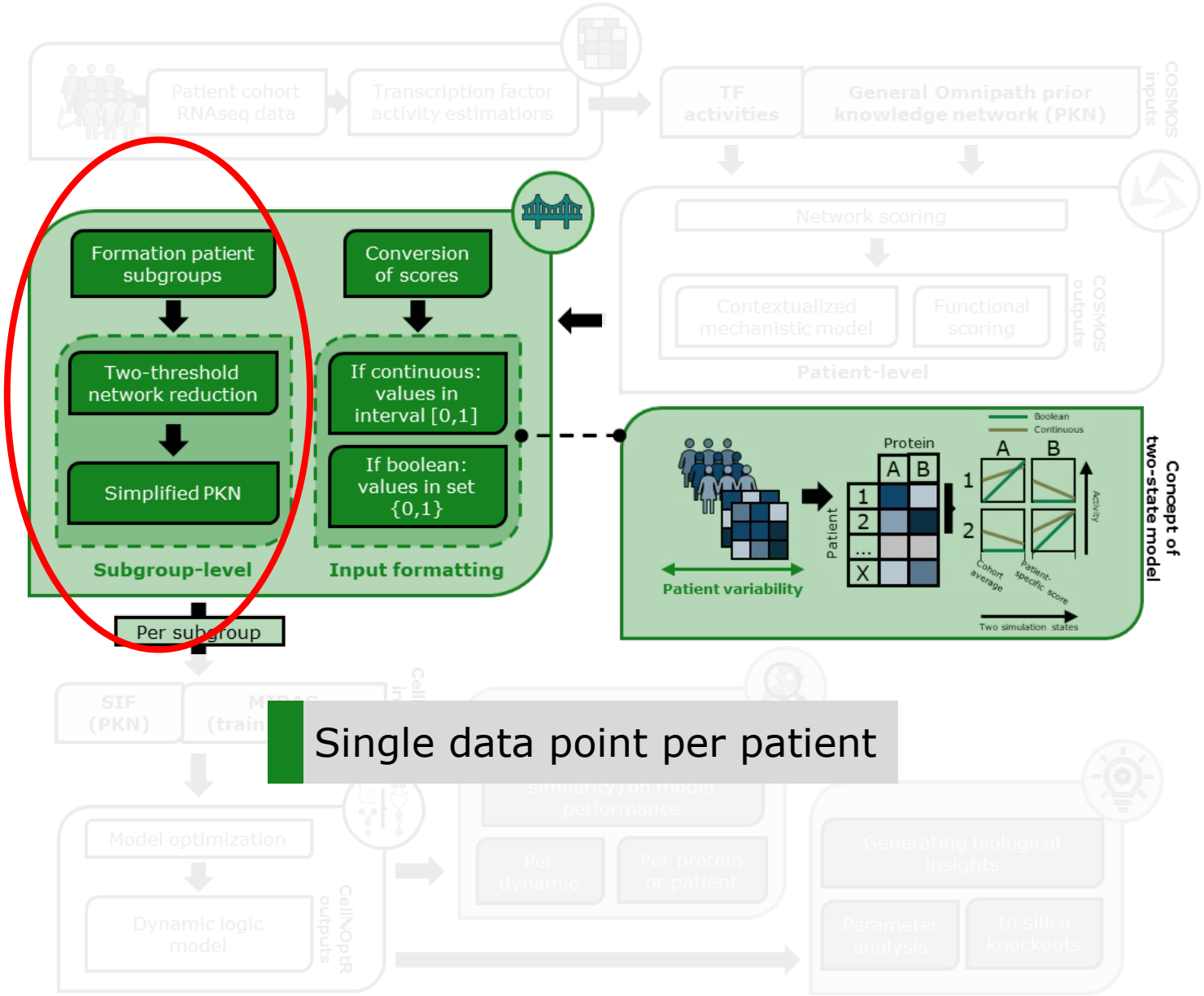
PKN : union of patient-specific MOON networks

Union of networks with thousands of edges results in a very large PKN

Two-threshold network reduction

- Get most relevant proteins
- Retain network structure

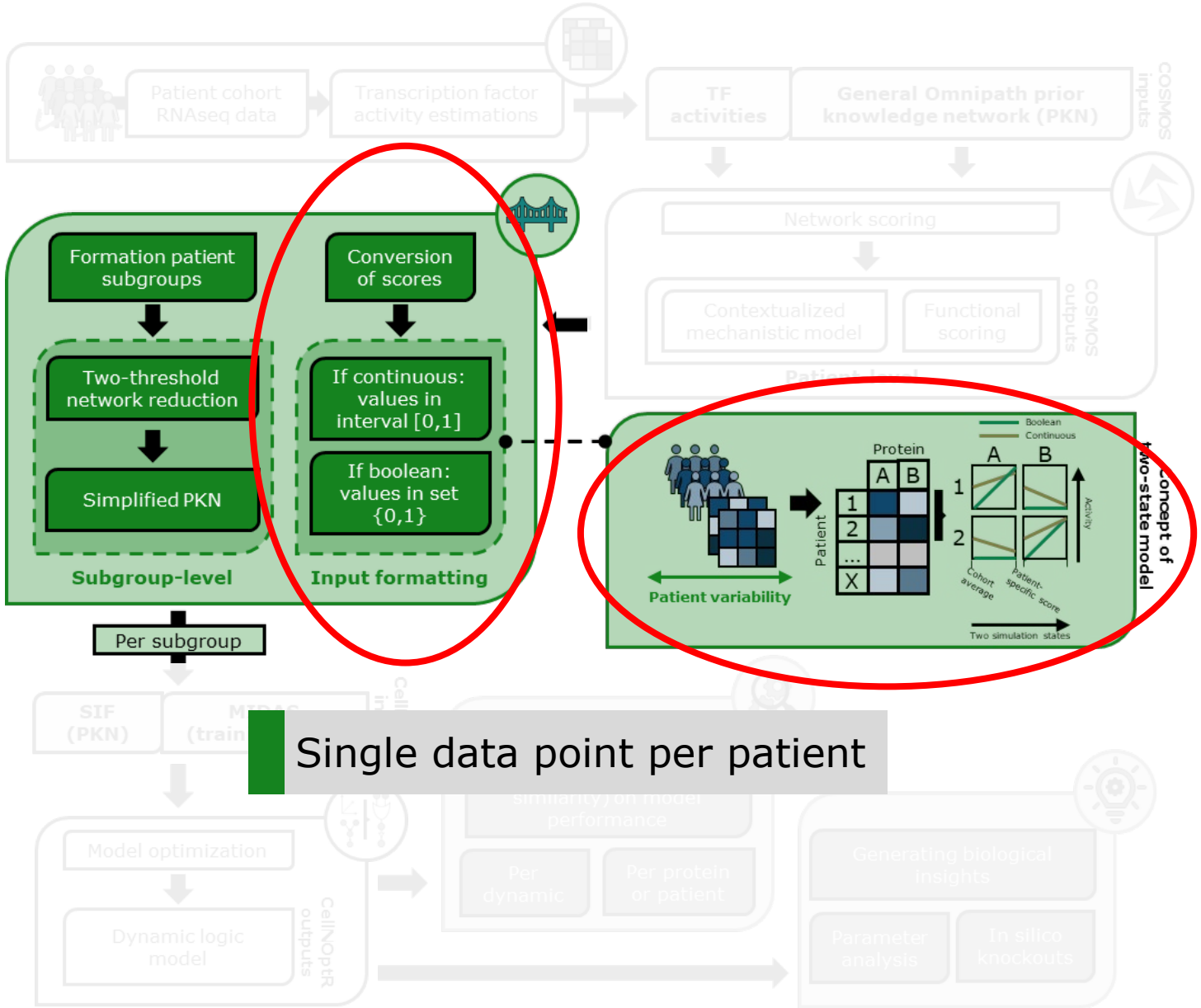




Single data point per patient

FICUS

- Overview of the main tools (COSMOS, CellNOpt)
- Preparation of PKN
- Preparation of training data
- Post-hoc analyses



Single data point per patient

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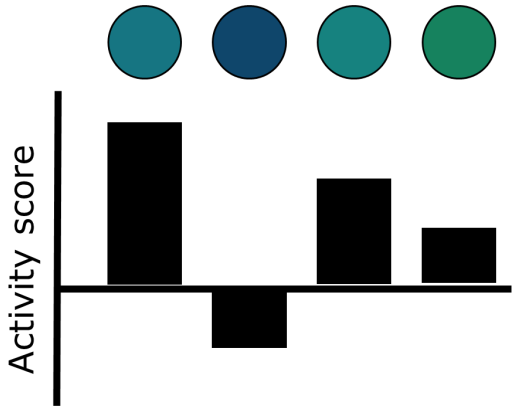
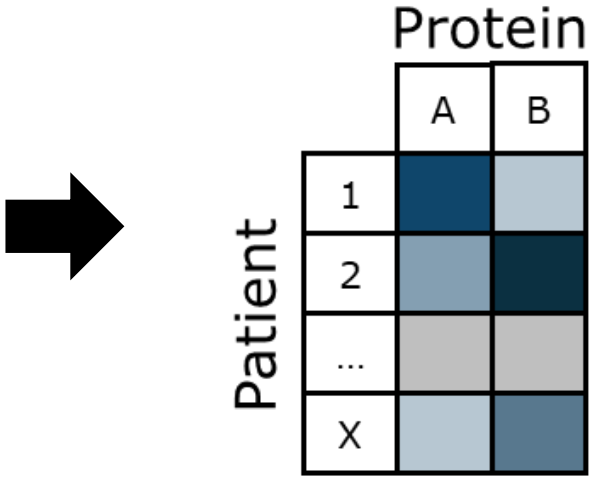
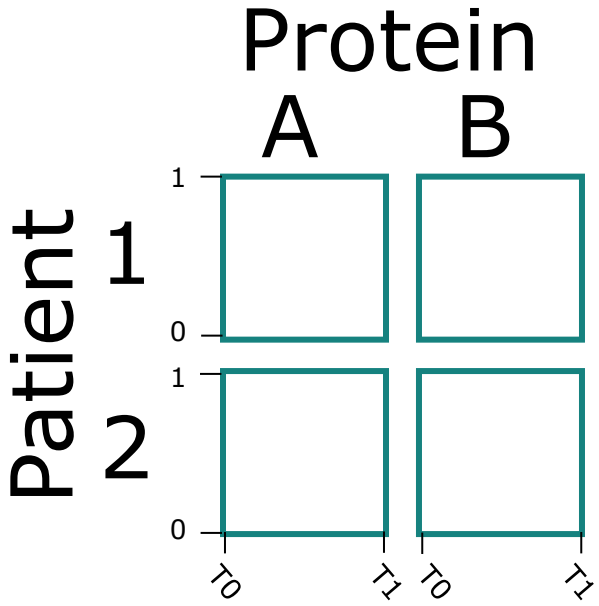


Functional scores from MOON

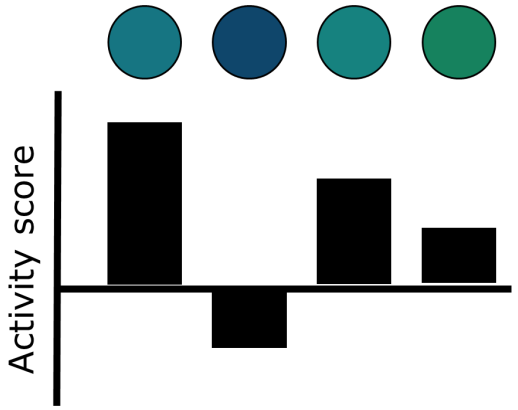


Training data

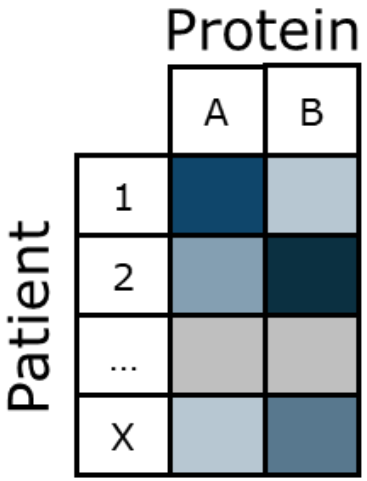


Logic-ODE simulations

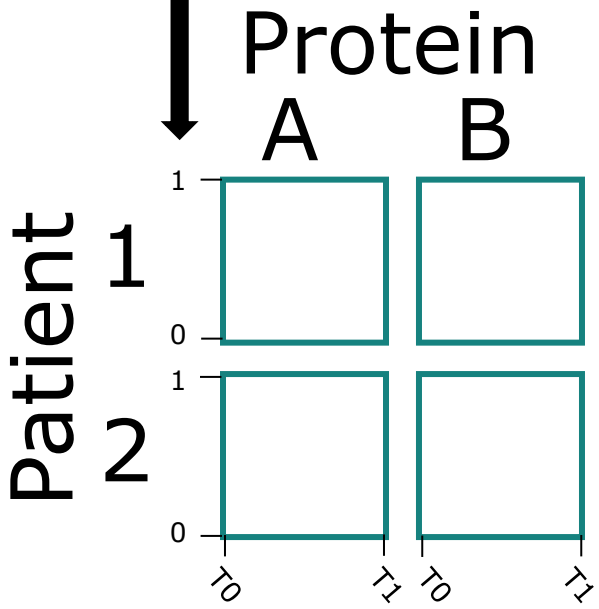
Y values : Rescaled functional scores



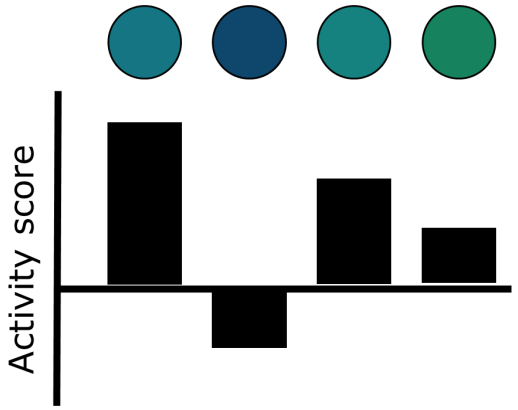
Functional scores from MOON



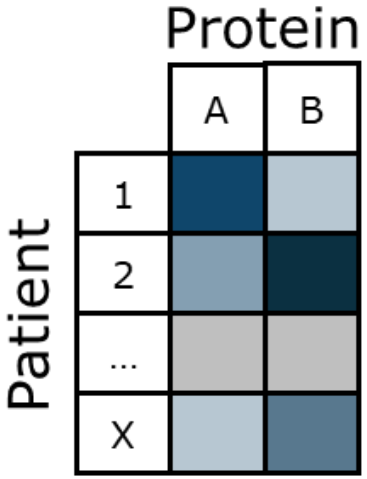
Training data



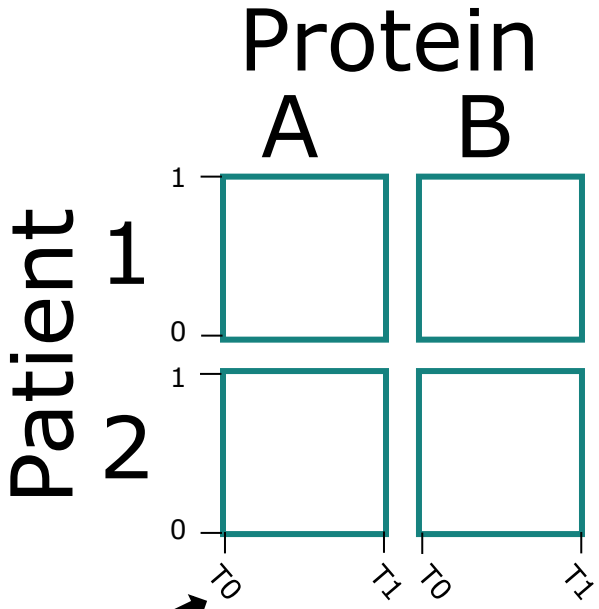
Logic-ODE simulations



Functional scores from MOON

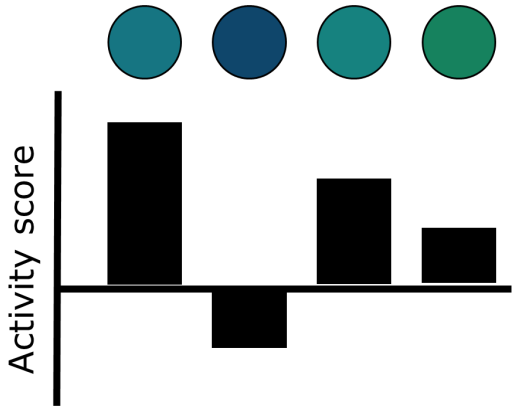


Training data

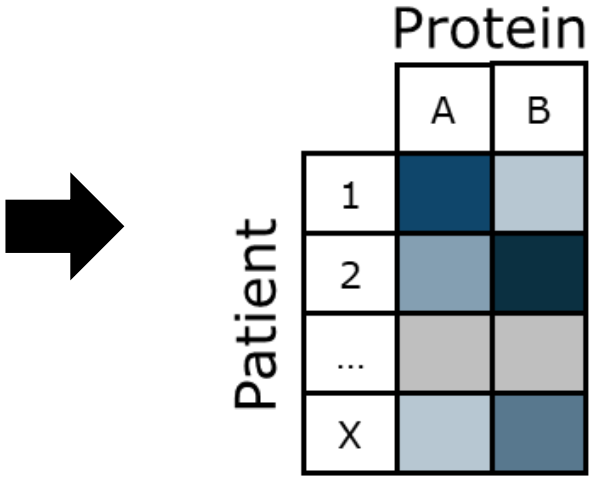


X values : Time…?

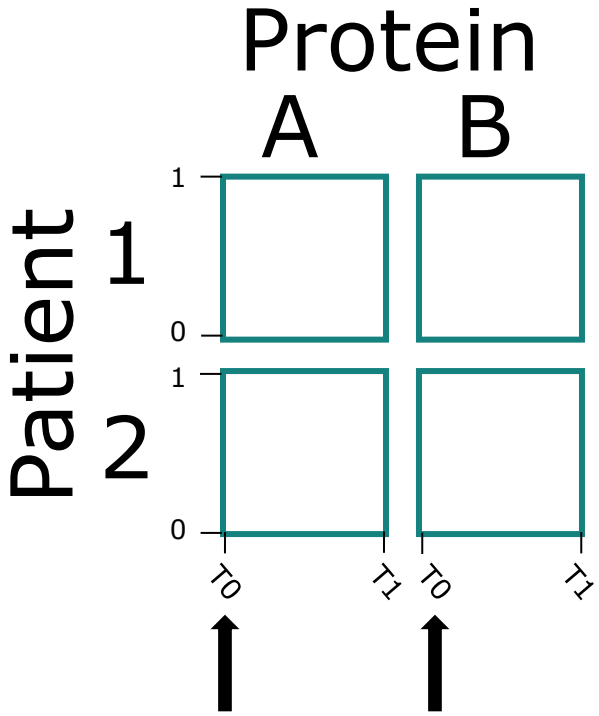
Logic-ODE simulations



Functional scores from MOON

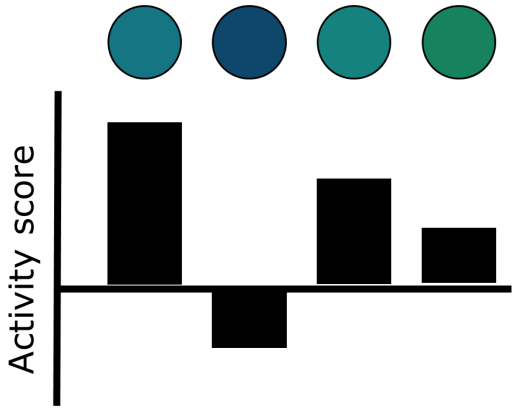


Training data

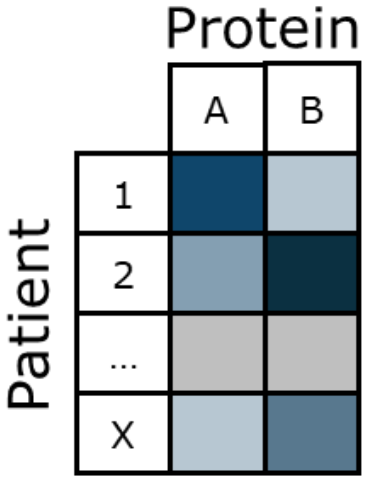


Logic-ODE simulations

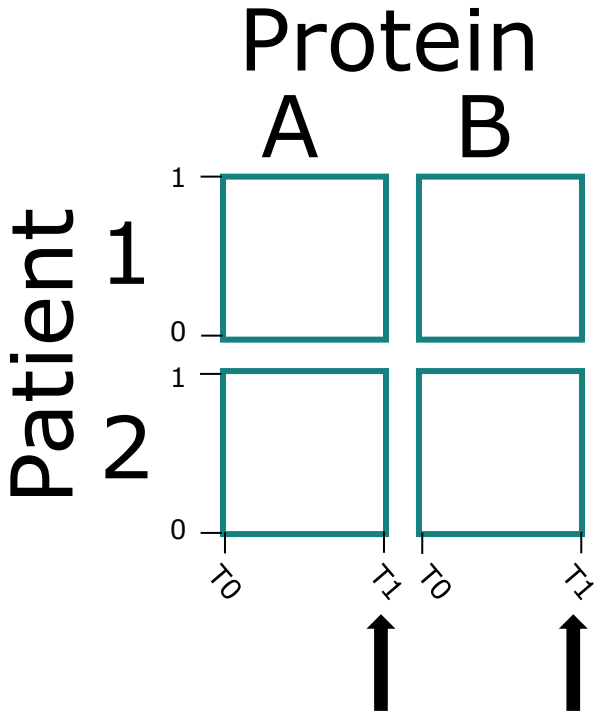
T0 : same starting point (baseline)



Functional scores from MOON

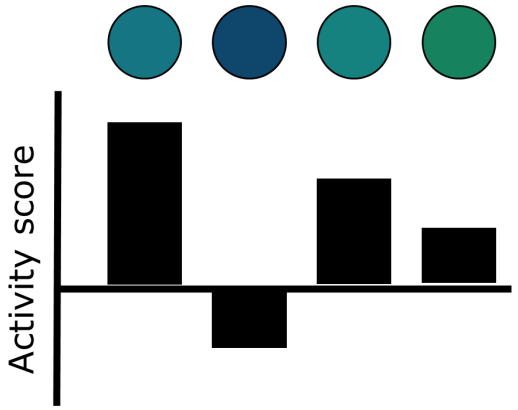


Training data

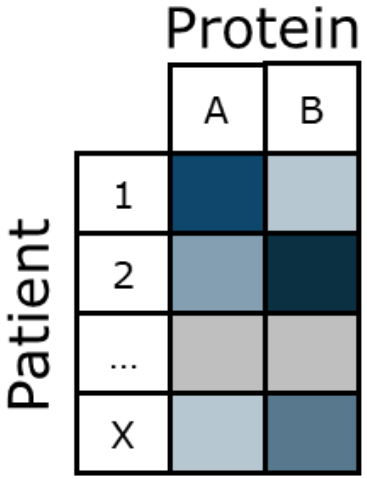


Logic-ODE simulations

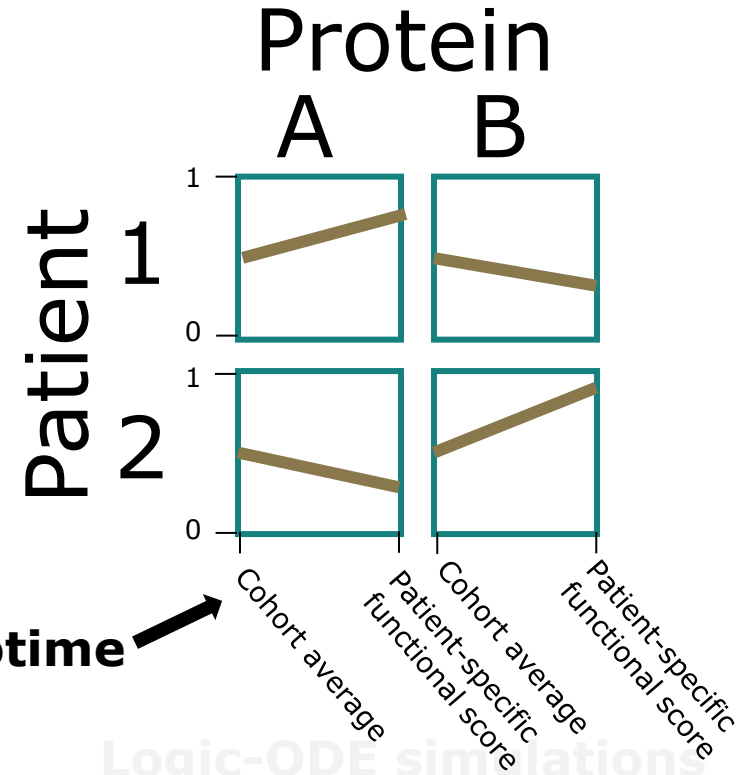
T0 : same starting point (baseline)
T1 : represents patient-specific responses



Functional scores from MOON



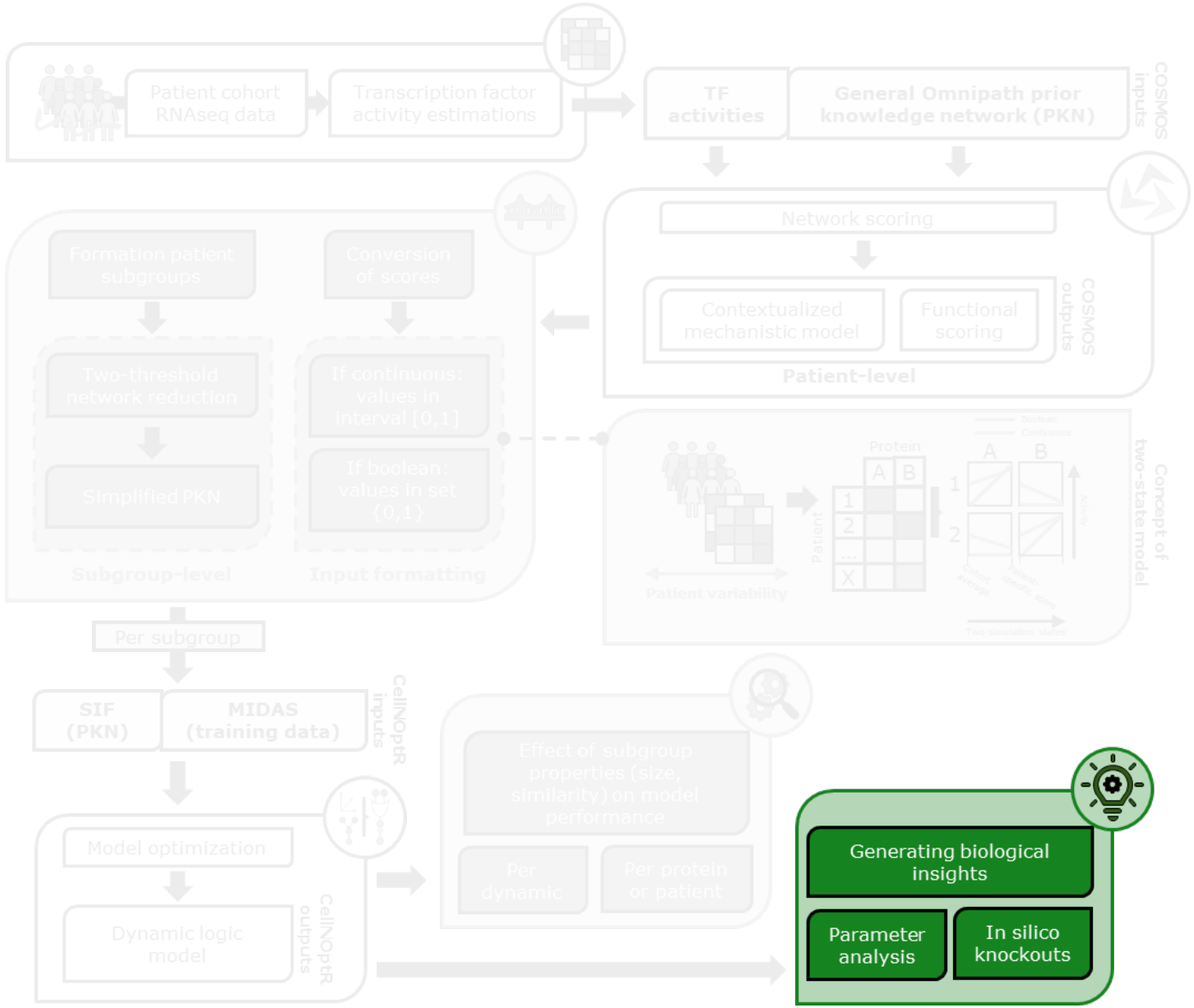
Training data



X values : Pseudotime

Logic-ODE simulations

T0 : same starting point (baseline)
T1 : represents patient-specific responses



FICUS

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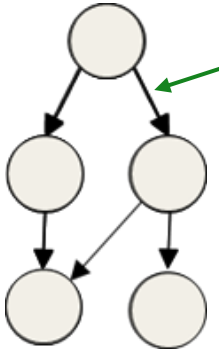
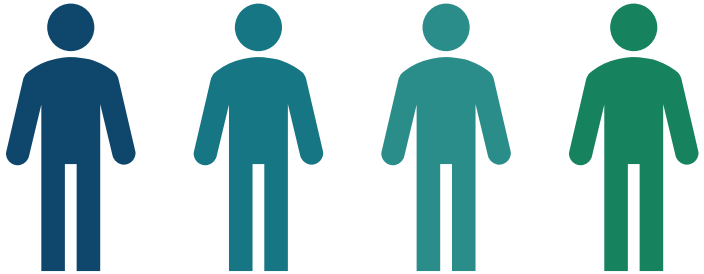
Prior knowledge network

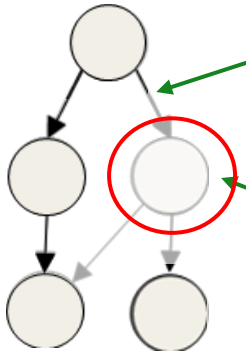


Assign k parameter for each interaction

Assign tau parameter for each protein

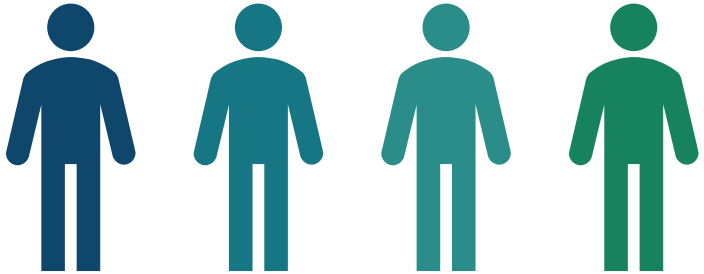


Prior knowledge network

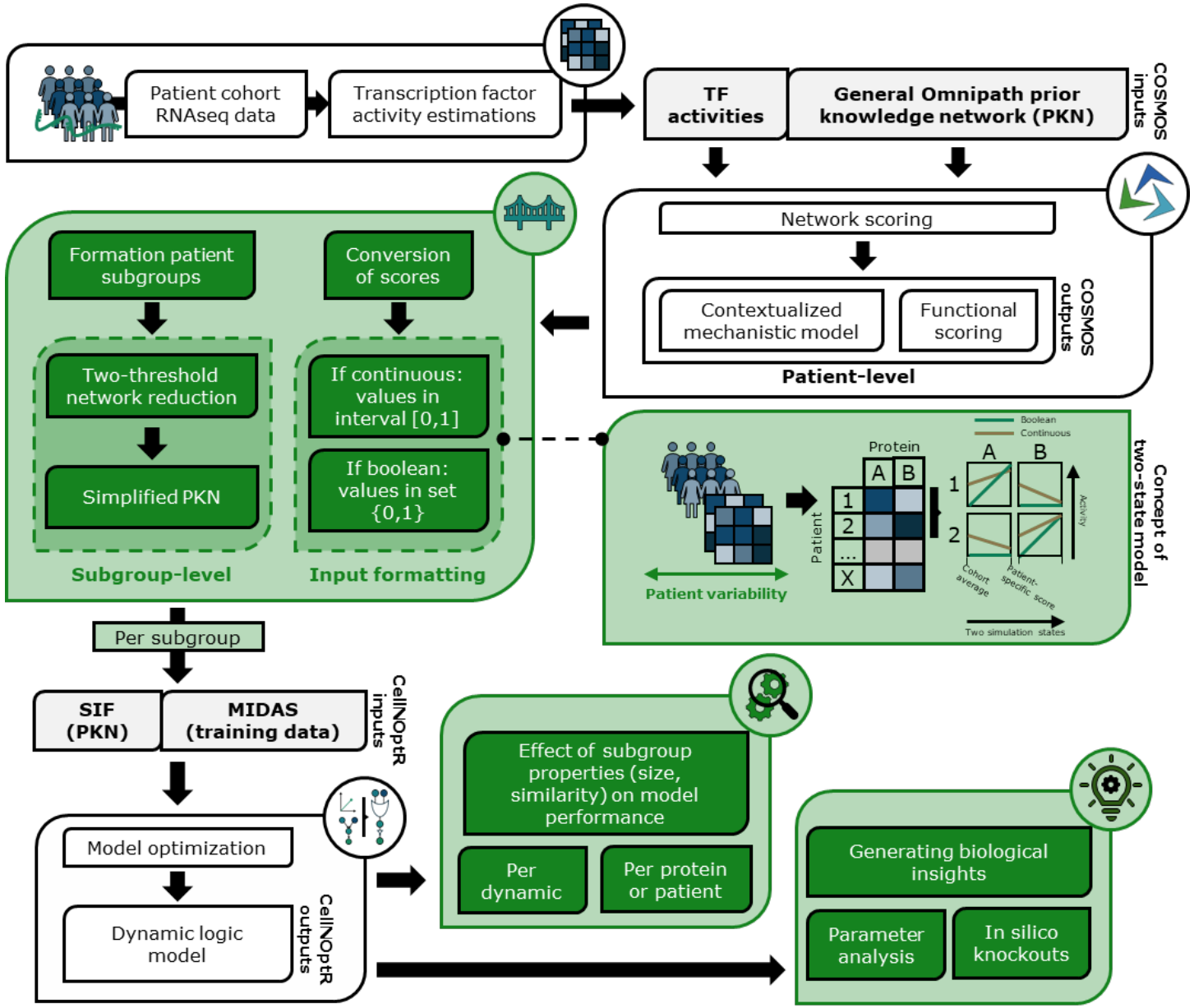


Assign k parameter for each interaction

Assign tau parameter for each protein



In silico knockout



FICUS

- Overview of the main tools (COSMOS, CellNOpt)
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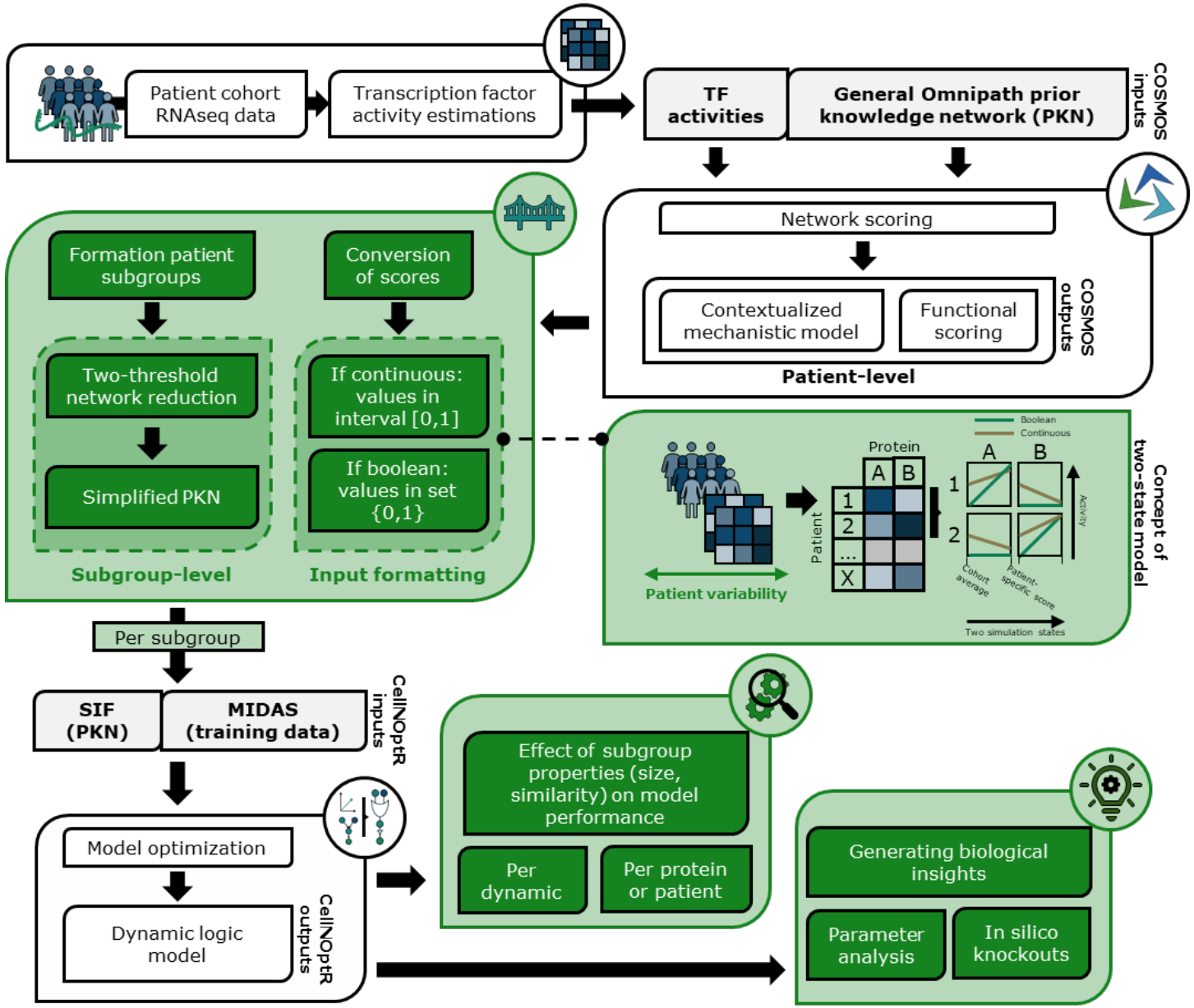
After break:
Examples
Introduction to practical



FICUS

- Overview of the main tools (COSMOS, CellNOpt)
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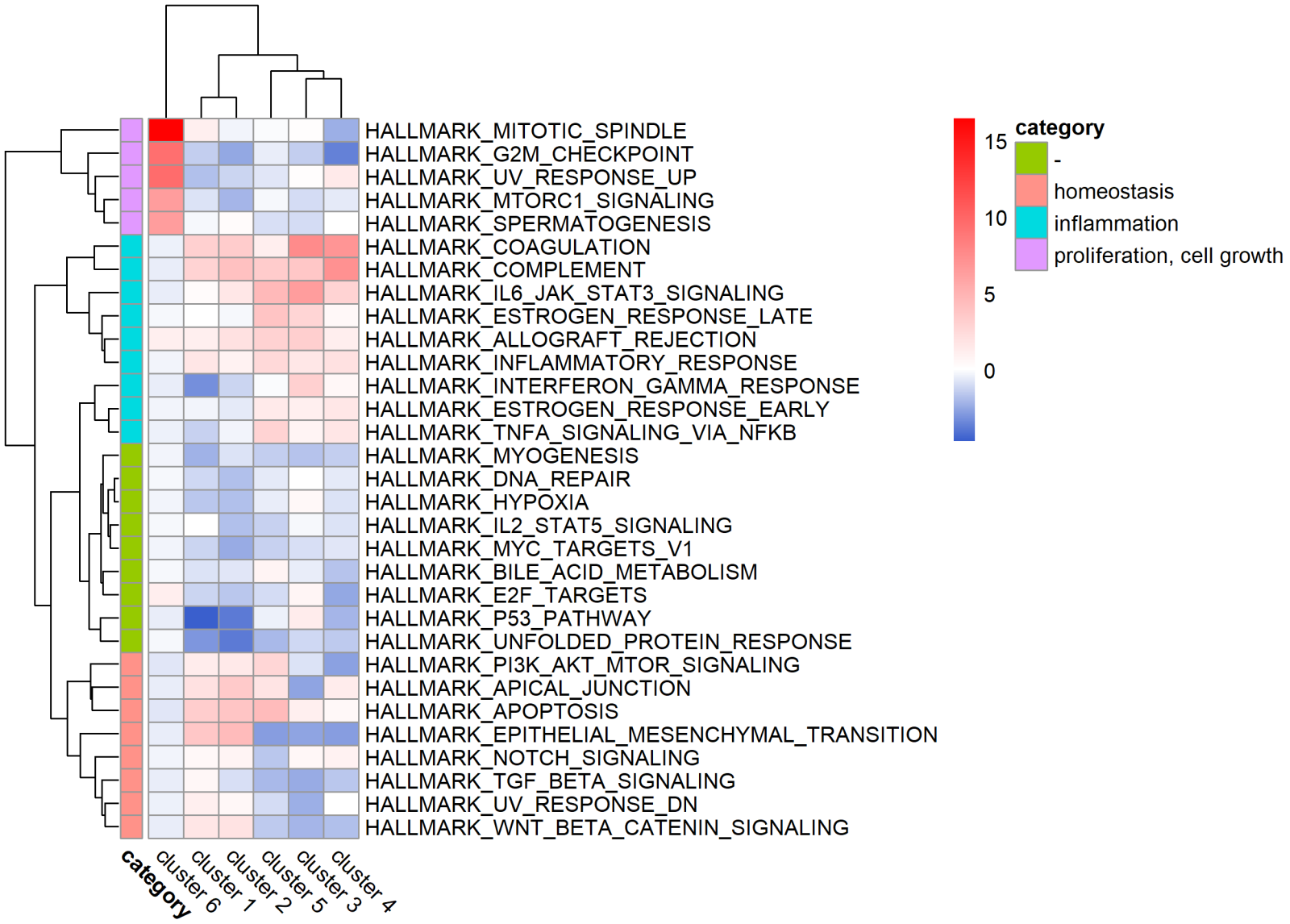
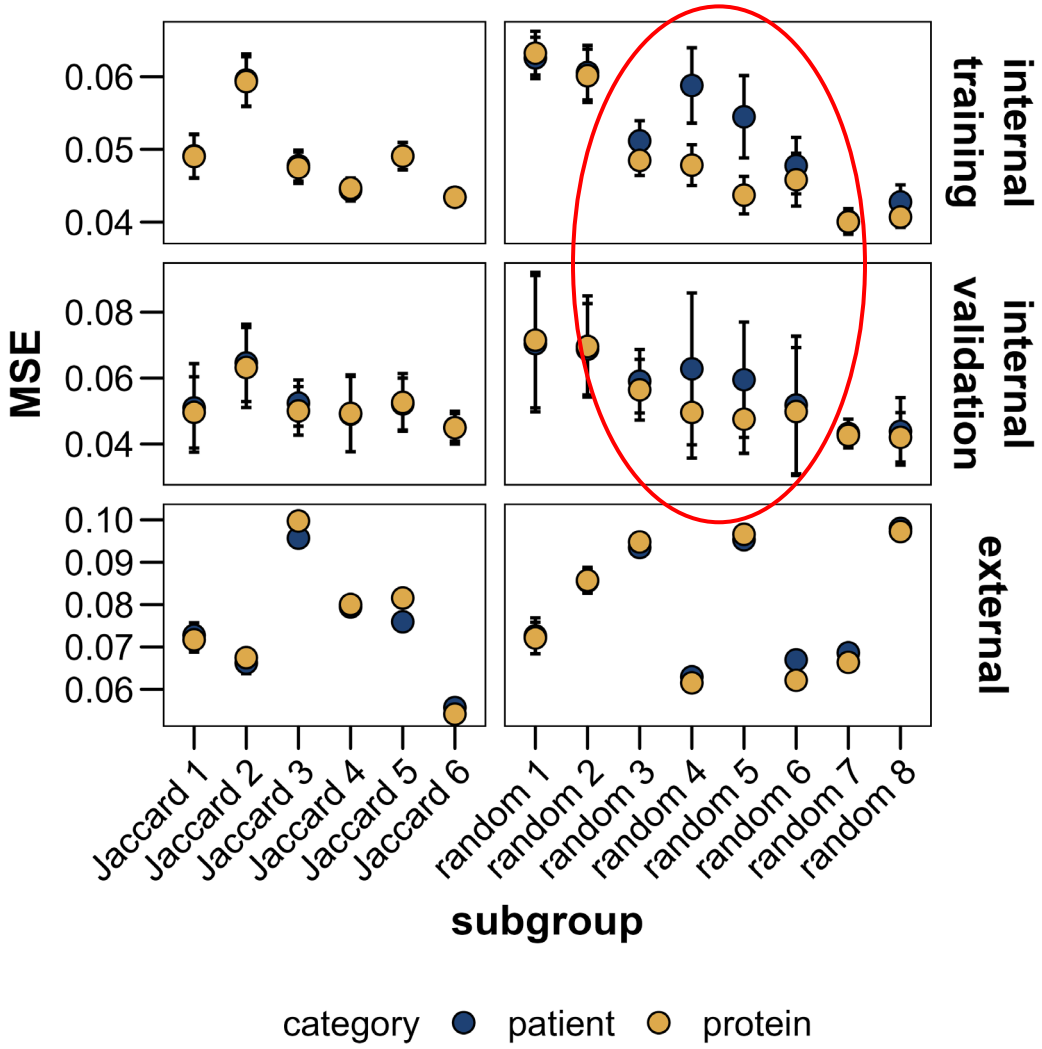
Subgroups can be further analyzed with enrichment analyses



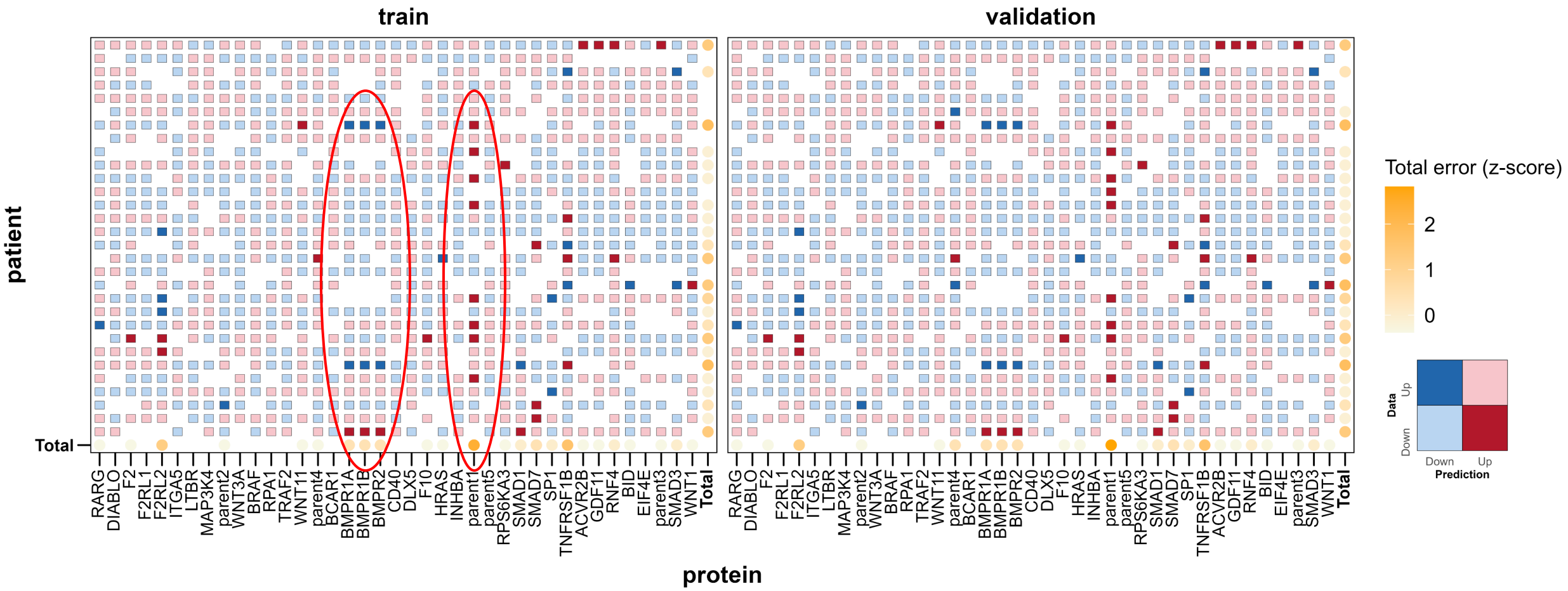
SU2C-MARK lung cancer cohort

Error per protein or per patient highlight the advantage of hierarchical clustering over random subgroups



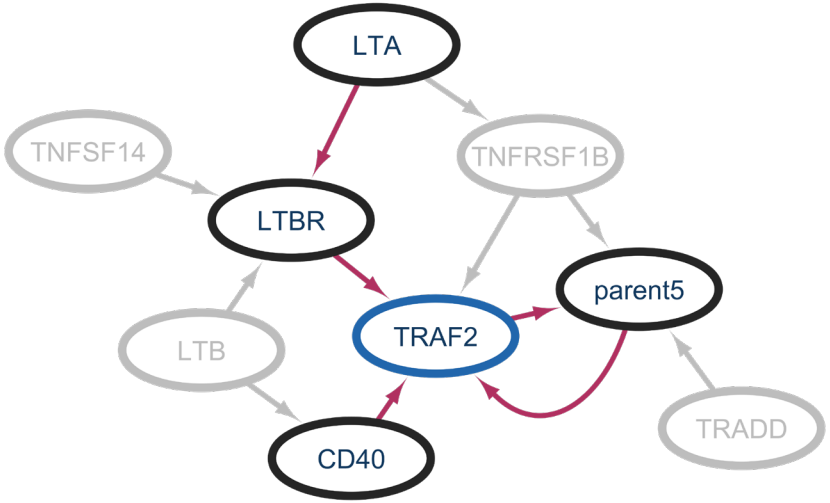
SU2C-MARK lung cancer cohort

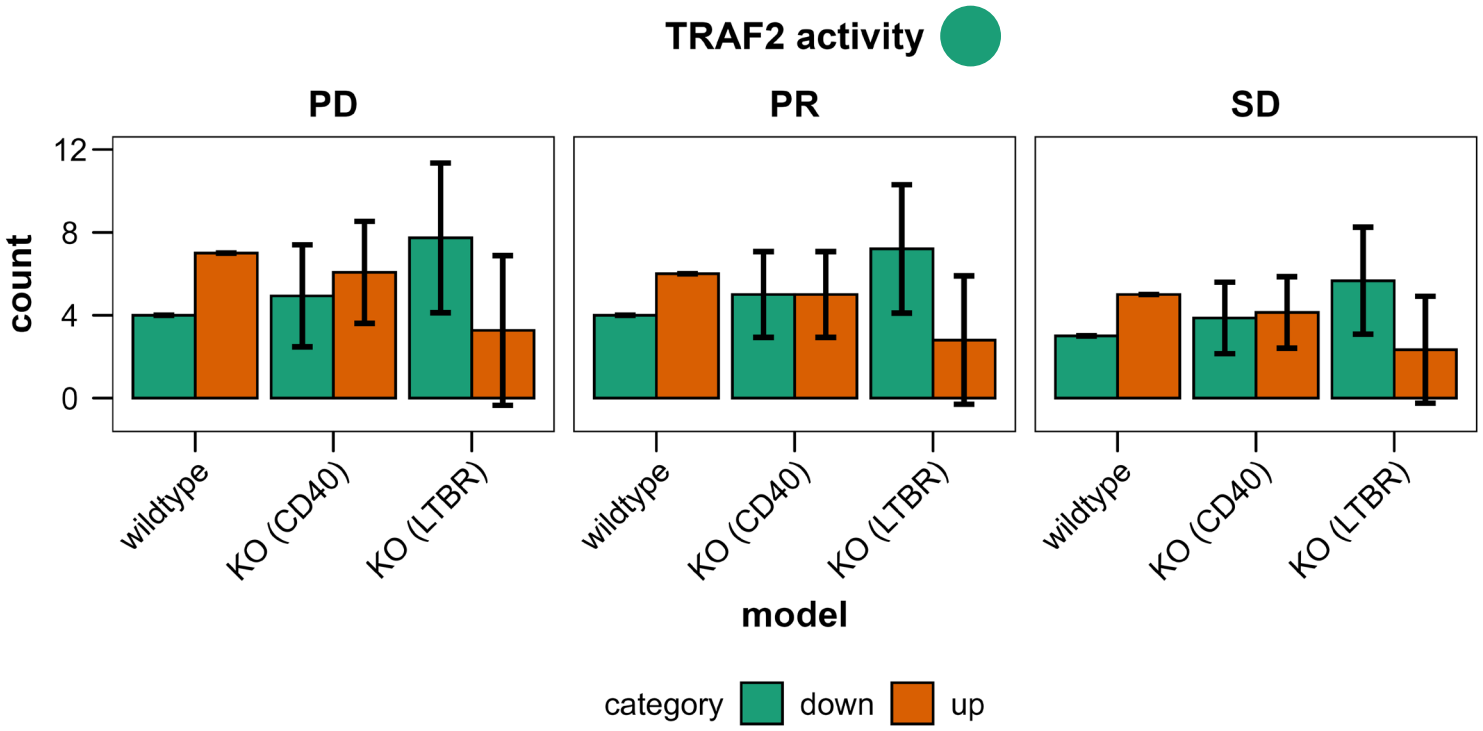
In-depth error per protein or per patient can reveal areas of interest for follow-up analyses



SU2C-MARK lung cancer cohort

In silico screenings can be valuable for identifying potentially effective targets for knockouts

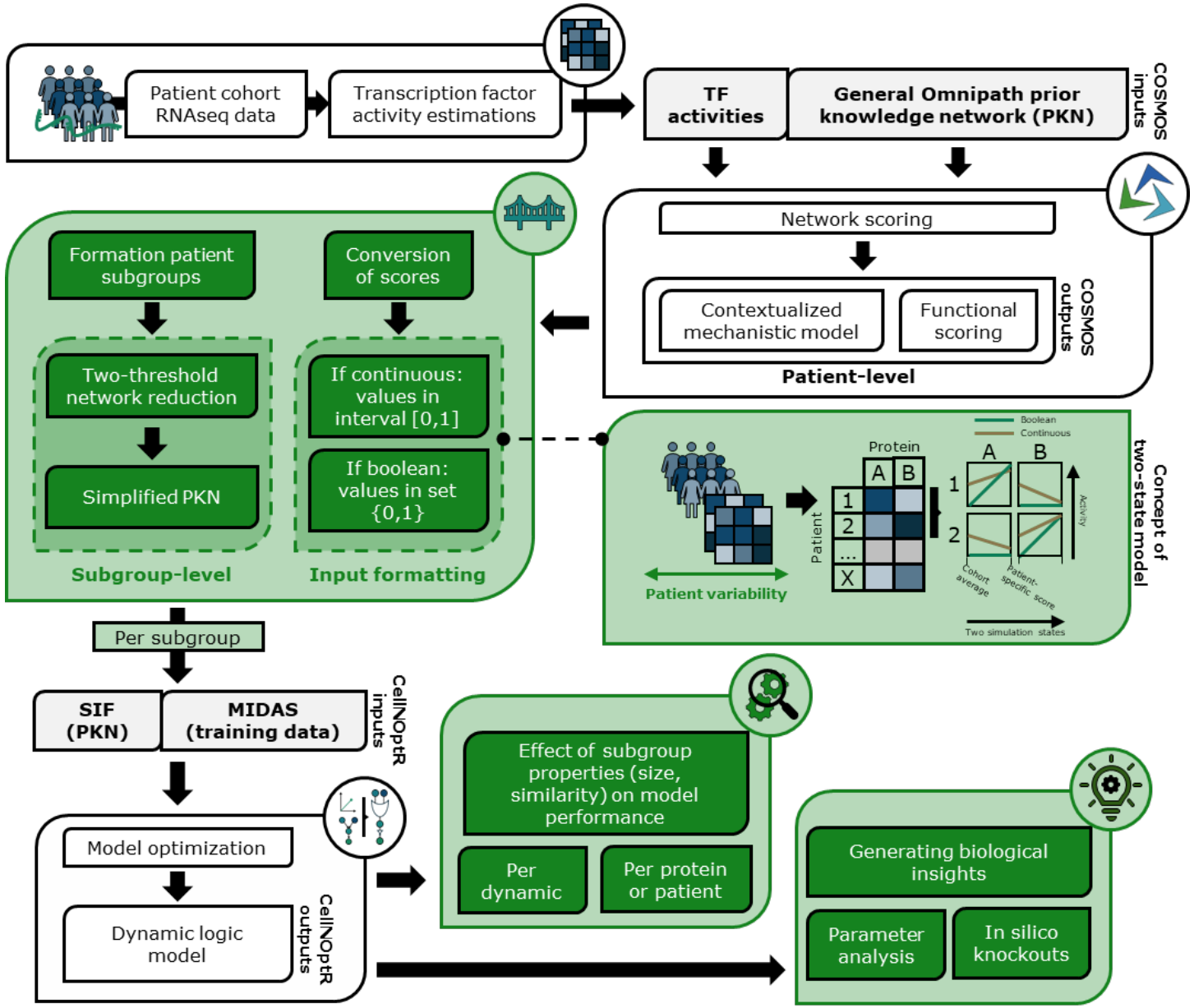




Which proteins have a significant impact on TRAF2?

Can we use knockouts to change non-responder's profile to a responder's profile?

During this session…



FICUS

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During this session…

COSMOS

Learning outcomes

- Deriving **patient-specific prior knowledge networks** from **bulk transcriptomics** including network simplification

Subgroups to fit logic models to omics data with one data point per patient

- **Stratifying patients** based on protein networks

- Converting COSMOS outputs to CellNOpt compatible inputs, **transforming static protein representations into dynamic mechanistic models**

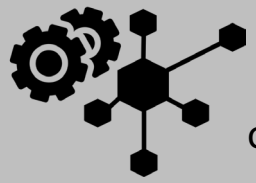
Two-state pseudotime approach and network reduction

CellNOpt

- Using logic-ODE models to perform **in silico knockout screenings**

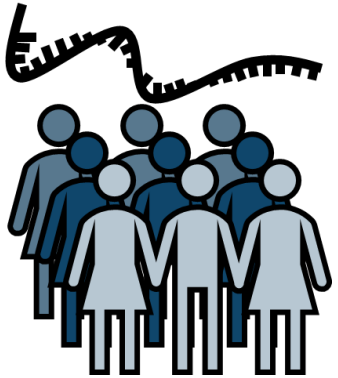
Generating biological insights and / or target screening

During this session…

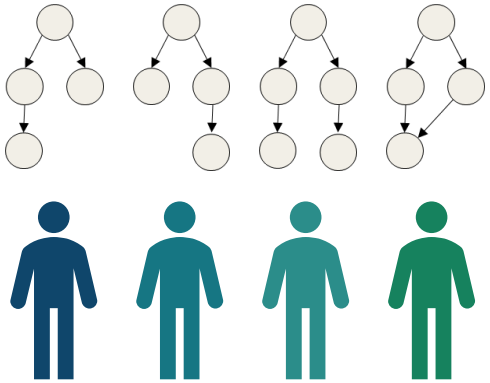


Unraveling signaling dynamics through computational models

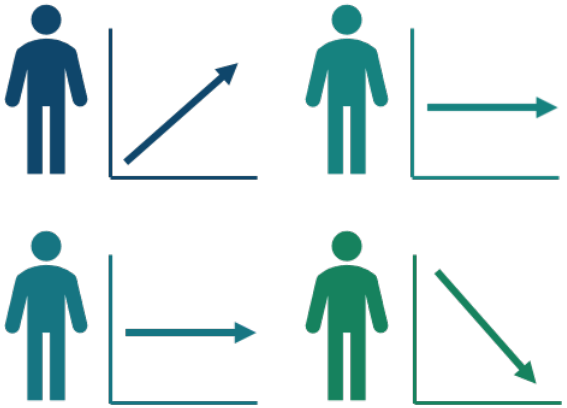
Omics dataset

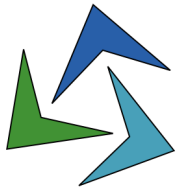


Static networks

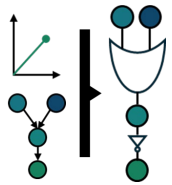


Dynamic models





COSMOS (Dugourd et al., 2024)



CellNOpt (Terfve et al., 2012)

From omics data to networks to mechanistic models

From omics data to networks to mechanistic models

EMBL-EBI Course

Systems biology: from large datasets to biological insight

15-05-2026

Bi-rong Wang (PhD Candidate at Biomedical Engineering, TU/e)